MILITARY SPECIFICATION

CONTAINER CRANE, ROUGH TERRAIN, WHEEL-MOUNTED, HYDRAULIC, DIESEL-POWERED

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for diesel-engine-driven, four-wheel drive, wheel-mounted, rough terrain, pneumatic-tired, hydraulic container cranes.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photo-sensitive Anodized Aluminum.
QQ-P-35	- Passivation Treatments for Corrosion Resisting Steels.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
RR-W-410	- Wire rope and strand.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950

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MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type,
MIL-P-14105	- Paint, Heat Resistant (for Steel Surface).
DOD-P-24555	- Paint, Aluminum Heat Resisting (650°C), Low Emissivity.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Glove Set, Chemical Protective.
MIL-F-43987	- Footwear Cover, Chemical Protective (Overboots).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-M-51282	- Mask, Chemical - Biological, field, M17A1.
MIL-H-51291	- Hood, Chemical - Biological Mask: M6A2.
MIL-P-52192	- Primer, Coating, Epoxy.
MIL-A-52363	- Air Cleaners, Intake: Dry-Type (for Internal-Combustion Engine)
MIL-H-52471	- Hose and Hose Assemblies, Rubber: Hydraulic Pressure-Type, General Specification for.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-T-62314	- Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE).

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STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards For Federal Services.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1366	- Material Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS51113	- Switch, Vehicular Lights, 24-Volt DC, Waterproof.
MS51117	- Pintle Assembly, Towing: 100,000 Lbs. Capacity, Manual Release.
MS51318	- Headlight: Blackout, 24 Volt, Waterproof.
MS51330	- Stop Light - Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS52125	- Composite Light - Tail, Stop, Turn and Marker.
MS52126	- Composite Light - Front, Turn, Park and Marker.
MS52131	- Connector, Plug, Electrical Intervehicle Power Cable.
MS53052	- Bracket Assembly, Liquid Container, Five Gallon.
MS500004	- Pad Eye Vehicular Towing Lug.
MS500048	- Towbar, Motor Vehicle, Light, Medium and Heavy Duty.

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HANDBOOK

MILITARY

MIL-HDBK-157 - Transportability Criteria.

(Copies of military specifications and military standards required by manufacturers in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Military handbooks should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

TACOM

12258941 - Connector, Receptacle 54 Pin.
12258955 - Vehicle Test Card.

REPORT

CR 82-588-003 - STE/ICE Design Guide for Vehicle Diagnostic Connector.

(Application for copies should be addressed to Department of the Army, U.S. Army Tank-Automotive Command, ATTN: DRSTA-RGD, Warren, MI 48090.)

UNITED STATES DEPARTMENT OF AGRICULTURE (USDA)

Forest Service Standard 5100-l.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Protection of Environment.
Title 40 Code of Federal Regulations.

DEPARTMENT OF LABOR

Occupational Safety and Health Administration (OSHA) Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the issue of the DoDISS specified in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

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B93.30M Hydraulic Fluid Power - Contamination Analysis Data - Reporting Method.
Z53.1 Safety Color Code for Marking Physical Hazards.
MH11.3 Load Handling Symbols for Powered Industrial Trucks.
Z26.1 Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.
B30.5 Mobil and Locomotive Cranes.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D - 1171 Rubber Deterioration - Surface Ozone Cracking Outdoors or Chamber.

D - 3951 Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Sections 5 and 6, Welding Qualifications.
D14.3 - Earthmoving and Construction Equipment Welding, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

Standard 114 - Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL FLUID POWER ASSOCIATION (NFPA)

T.3.9.18 - 1976 - NFPA Recommended Standard for Calibration of Liquid Automatic Particle Counters.

(Application for copies should be addressed to the National Fluid Power Association, P.O. Box 49, Thiensville, WI 53092.)

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POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

Mobile Hydraulic Crane Standards (PCSA Standard No. 4)

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700 111 E. Wisconsin Ave., Milwaukee, WI 53202).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited in 2.1.1, 2.1.2, and 2.2, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The crane shall be four wheel drive, pneumatic tired, diesel engine driven, wheel-mounted, full revolving hydraulic telescoping boom, container crane. The crane shall conform to OSHA, ANSI, and PCSA standards as applicable.

3.1.1 Rated load. Rated load (see 6.8.1) for the crane shall be as specified by the load rating chart for the required radius.

3.1.2 Physical characteristics.

3.1.2.1 Weight. With the crane fully fueled and serviced, the axle weight shall not exceed 59,000 pounds for the heaviest axle. Maximum allowable vehicle weight shall be 111,000 pounds.

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3.1.2.2 Overall height. The overall crane height, with the boom horizontal shall not be more than 13 feet.

3.1.2.3 Axle spacing. Axle spacing for the crane shall be not less than 12 feet.

3.1.2.4 Width. The width of the crane including tire bulge at manufacturer's recommended tire pressure shall be not more than 12 feet 6 inches.

3.1.2.5 Ground clearance. The ground clearance and the angles of approach and departure of the crane as defined in SAE J1234 shall not be less than 14 inches and 13 degrees respectively. The crane shall be fully fueled and serviced. The removal of the outrigger floats are allowed.

3.1.3 Transportability. The crane shall be transportable without damage, by highway, marine, and rail transport modes. Transportability criteria are established by MIL-STD-1366 and MIL-HDBK-157 as described herein.

3.1.3.1 Highway transportability. Disassembly of the crane is acceptable to meet height, width, and weight requirements specified herein. Disassembly shall be accomplished in not more than 12 clock hours, and reassembly of the crane shall be accomplished in not more than 12 clock hours by six persons using suitable material handling equipment as required. Disassembly/assembly shall not include cutting or welding. Slinging and tiedown provisions conforming to MIL-STD-209 shall be furnished on each assembly, (except tires and wheels), required to be lifted by crane on to the semi-trailer (M747). The limitations for highway transport are:

Height	- When mounted on the 44-inch high semi-trailer bed the maximum height above ground shall not exceed 162 inches.
Width	- The maximum width for highway transport shall not exceed 144 inches.
Weight	- The maximum payload weight for highway transport shall not exceed 60,000 pounds.

3.1.3.2 Rail transportability. All components removed for highway transport shall also be removed for rail transport. The crane and the removed components shall withstand railroad humping without permanent deformation or damage to any of its components when tested in accordance with MIL-STD-810, Method 516.3. Procedure VIII - Rail Impact.

3.1.3.3 Marine transportability. The crane without disassembly, shall be transportable by breakbulk cargo ships, roll-on/roll-off (RORO) ships, C-8 and larger, lighter aboard ship (LASH), barge carrying ships (SEABEE), LARC60 and larger amphibious vessels (LCM8, LCU-1466, LCU-1646, BDL MARK I, Bk 7001, BC 7005, and BC 231-A) in accordance with MIL-HDBK-157 and MIL-STD-1366.

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3.1.4 Crane structure. The structure of the crane shall be such that stresses induced during required operations do not exceed the allowable strength margins of SAE J1063.

3.2 First article. When specified (see 6.2), 4 or more cranes shall be subjected to first article tests and inspection as specified in section 4. Any changes or deviations of cranes from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish cranes conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. It is encouraged that recovered material be used when practical as long as it meets the requirements of this specification (see 6.8.7).

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.3.1.1 Material deterioration and control. The crane shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may affect the normal performance of the crane or during storage of the crane when stored in accordance with proper storage procedures for the type of storage involved.

3.3.1.2 Drain holes. Drain holes shall be provided where applicable, to prevent collection or entrapment of water or other unwanted fluids.

3.3.2 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.3.3 Tire age. Tires shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance o£ the crane by the Government.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the crane may be newly fabricated from recovered materials to the maximum extent practicable, provided the crane produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the crane.

3.4 Environmental and use requirements.

3.4.1 Operating conditions. The crane shall withstand the usage encountered in military operations without damage, permanent deformation or failure of components. The military operations include

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loading and unloading 20-foot, 35-foot and 40-foot standard ANSI and ISO containers from railroad cars and semitrailers. Operations also include loading and unloading slung palletized projectiles on ammunition transporters. These operations shall be capable of being conducted over unprepared, unstabilized surfaces, especially over rough terrain such as beach operations in soft sand, snow and mud where high flotation is needed, and travel without load between work sites.

3.4.2 Operating temperatures. The engine shall start without preheating within 5 minutes and the crane shall operate as specified herein within 15 minutes after engine start in any ambient temperature in the range of -25° F to +120° F.

3.4.3 Storage temperature. The crane shall not be damaged by storage under the conditions indicated in table I.

TABLE I. Storage.

Air Temperture	Time Period
Plus 160° F	1 hour daily.
Plus 155° F	4 hours daily.
Minus 25° F	6 hours daily.

3.4.4 Rain. The crane shall start within 5 minutes and operate as specified herein when tested as specified in 4.6.2.23.

a. Without malfunction of any electrical component, switch, gage, light or instrument.
b. Without leakage of water into filters incorporating a sediment chamber in the air-intake filter in excess of 8 percent of the fluid capacity of the sediment chamber.
c. Without leakage of water into the torque converter oil chamber, engine crankcase sump, transmission, transfer case, instruments, switches, gages, lights, or fuel tank.
d. Without any detrimental leakage into the cab.

3.5 Safety. The crane shall meet the safety requirements specified in OSHA 29CFR 1926, PCSA 4, and ANSI B30.5 unless otherwise specified herein. All rotating and reciprocating parts and parts subject to high temperature shall be guarded when such parts are exposed to contact by operator and maintenance personnel performing daily maintenance functions. Nonfunctional edges shall be rounded, projecting points shall be blunted or rounded, and excessive length of fasteners shall be avoided. Steps and platforms for entering the crane shall have antiskid, ice resistant surfaces. Oil level indicators for engine, transmission, and other components shall be within easy and safe access of the operator or maintenance personnel. Portions of the superstructure that extend out and over the carrier when being swung where it may cause a striking or crushing hazard for

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personnel, shall have the lower and back area taped in color coding in accordance with ANSI Z53.1.

3.5.1 Seatbelt. An adjustable weather-resistant seatbelt with buckle conforming to SAE J386, type I shall be furnished equipped with sleeves and retractor or seat mounted.

3.5.2 Backup alarm. A backup alarm conforming to SAE J994, type C or D shall be furnished. The backup alarm signal shall be audible above, and distinguishable from the surrounding vehicle noise level. The backup alarm shall be automatically disengaged in the blackout mode.

3.5.3 Load rating chart. Load rating chart in accordance with ANSI B30.5 shall be provided and shall contain not less than the following information:

a. Rated loads in pounds in not more than five-foot increments for the full range of operating radii with and without outriggers.
b. That the rated loads do not exceed 85 percent of the tipping loads on outriggers or 75 percent of the tipping loads on rubber.
c. Definition of the load to be lifted.
d. Hook block reeving.

3.6 Human factors. The characteristics of the fully assembled crane shall provide for operation and maintenance by personnel ranging from the 5th percentile female, through the 95th percentile male, arctic clothed, in accordance with SAE J833 and SAE J925. Components and equipment removed disassembled or relocated for operation maintenance, or transportability as specified in 3.1.3, which must be lifted or carried by operating personnel shall not exceed 50 pounds in weight for one person lifts and 100 pounds for two or more person lifts, and shall be provided with suitable handles or grasp areas. Applicable SAE human factors specifications shall be incorporated in the crane.

3.6.1 Noise limits.

3.6.1.1 Personnel limits. The noise produced by the crane shall conform with MIL-STD-1474 requirements at the operator's position when tested in accordance with 4.6.2.22. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5,1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 82dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.6.1.2 Exterior noise. The noise level of the crane (excluding horn) shall not exceed 88dB(A) when tested in accordance with SAE J88.

3.7 Maintainability. Provision shall be made for adjustment, servicing or replacement of parts and components. When access

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openings are used on the exterior of the crane, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover whenever it is required. Dimensions of hand or arm access openings shall be in accordance with SAE J925 for arctic clothing. When tested as specified in 4.6.2.30, the crane shall have a maintenance ratio (MR) of not more than 0.25. The MR is defined as the ratio of total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. Scheduled maintenance shall be required not more often than every 50 hours, except for before-and-after operation checks. Each operation in the following maintenance operations list shall be accomplished by one person (except where otherwise specified) in not more than the time specified, using common tools and special tools, if any, furnished with the crane. Test points identified for connection of STE/ICE to measure functions in the Tk mode (see 3.41) shall be accessible without the removal of other major assemblies and installed attachments not normally removed.

a. Remove, replace and adjust all engine-driven belts - 3/4 hour.
b. Remove and replace alternator - 1/2 hour.
c. Remove and replace all hydraulic-system filters - 1 hour.
d. Remove and replace engine coolant-system hoses - 1 hour.
e. Drain engine lubricating oil, remove and replace oil filter elements and refill crankcase - 1 hour.
f. Remove and replace fuel filter elements - 1/2 hour.
g. Disconnect battery cables, remove and replace batteries, reconnect battery cables - 1/2 hour.
h. Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill-converter and transmission - 1 hour.
i. Remove and replace starter - 1 hour.
j. Bleed and adjust brakes and refill master cylinder (if applicable) - 2 men - 1 hour.
k. Remove and replace headlight, floodlight and taillight bulbs (not including blackout lights) - 1/4 hour each bulb.
l. Remove, disassemble, service and install air cleaner - 1/2 hour.
m. Lubricate all lubrication fittings with 2 strokes per fitting using the gun and extension specified in 3.29.1 - 1 hour.

3.8 Reliability. The crane(s) shall have a mean time between failure (MTBF) of 100 hours (see 6.8.9) when tested as specified in 4.6.2.29, using the failure definition in 4.6.2.29.1.

3.9 Performance. The crane shall conform to the requirements specified herein under all conditions specified without damage or permanent deformation. There shall be no evidence of accelerated wear or failures when operated under service conditions specified herein.

3.9.1 Stability.

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3.9.1.1 ANSI stability. The crane shall conform to ANSI B30.5 stability requirements.

3.9.1.2 Dynamic stability. The crane shall negotiate a 15 percent slope without load in full circle operation at maximum steer angles in both directions without any wheel leaving the ground. The crane while descending a 25 percent grade in the forward direction at not less than 2 mph, shall be capable of stopping within a distance of 20 feet by applying the service brakes at a pressure that prevents the rear wheels from leaving the ground. The crane, when stopped on the 25 percent slope, shall be held by the service brakes for not less than 1 minute.

3.9.2 Longitudinal gradeability. The crane shall ascend a 30 percent grade in forward gear range at a speed of not less than 2 mph without load.

3.9.3 Travel speed. The crane shall be capable of travel speeds on a level surface of not less than 18 mph forward without load.

3.9.4 Fording. The crane shall be capable of operation in seawater to a fording depth of 30 inches. Crane components shall be protected from damage, malfunction and corrosion due to immersion followed by drying for those components below the fording level, and due to splash and spray for those components above the fording level. When tested as specified in 4.6.2.26 the crane shall operate during and after fording operation without malfunction or damage from corrosion.

3.9.5 Lifting speeds.

3.9.5.1 Boom lift speed. The boom lift speed shall be not less than 1.5 degrees per second.

3.9.5.2 Hook lift speed. The main hook block lifting speed with the recommended number part lines shall be not less than 40 feet per minute. The auxiliary hook block lifting speed with the recommended number part lines shall be not less than 55 feet per minute.

3.9.6 Lowering speeds.

3.9.6.1 Boom lowering speed. The boom lowering speed shall be not more than 3.5 degrees per second and not less than 1.5 degrees per second at governed engine speed with the control valve in the full open position.

3.9.6.2 Hook lowering speed. The main hook block lowering speed with the recommended number part lines shall be not more than 80 feet per minute and not less than 40 feet per minute at governed engine speed with the control valve in the full open position. The auxiliary hook block lowering speed with the recommended number part lines shall be not more than 100

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feet per minute and not less than 60 feet per minute at governed engine speed with the control valve in the full open position.

3.9.7 Boom extension and retraction speeds. The crane shall be capable of extending the boom at a 60 degree boom angle without hook block load from the retracted boom position to full boom extension at a constant speed of not less than 35 feet per minute. The crane shall be capable of retracting the boom at the 60 degree lift angle from full boom extension to the retracted boom position at a speed of not less than 50 feet per minute with no hook block load.

3.9.8 Curb-clearance circle. The curb clearance circle as defined in SAE J695 shall be 52 feet, maximum.

3.9.9 Crane capacity. The crane on outriggers shall be capable of engaging and lifting, swinging 360 degrees, and disengaging the following loads plus the weight of the hook block at the specified heights and radii (see 6.8 for definitions):

Load	Radius	Hook Height
15,000 lb.	35 feet	20 feet
48,000 lb.	27 feet	30 feet
73,000 lb.	22 feet	40 feet

The crane shall also be capable of lifting the 48,000 pound load over the front at a distance of 16 feet forward of the vertical plane of the foremost part of the crane.

3.9.10 Drift. The boom lift assembly shall be capable of holding rated load at a 60-degree boom angle with full boom extension not less than 10 minutes with not more than 1 inch of cylinder travel. Specified drift rates shall be measured when hydraulic oil temperature is not less than 120° F.

3.9.11 Endurance. The crane shall be capable of completing the test specified in 4.6.2.29 with no failures using the endurance failure definition in 4.6.2.29.2.

3.9.12 High speed. The crane shall be capable of 4 hours of high speed travel when tested as specified in 4.6.2.27 without malfunction, damage, or permanent deformation, and without exceeding specified temperature limits for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11.1.1), universal joints (see 3.11) or hydraulic system (see 3.12).

3.10 Diesel engine. For the purpose of selecting the engine, the crane shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. A sleeveless engine is acceptable. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410.

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Unless otherwise specified herein, the engine shall meet all operational requirements and conditions specified in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits when operating in a 120 °F ambient. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer.

3.10.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the crane under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer.

3.10.2 Engine accessories.

3.10.2.1 Starting priming system. Either a glow plug for each cylinder or an auxiliary fluid priming system, conforming to MIL-E-52649, type III, shall be furnished and shall be operable from the operator's seat. The size of the primer shall be in accordance with the engine manufacturer's recommendations. The primer may be mechanically or electromechanically operated. When the primer is mechanically actuated, it shall be furnished with a total cable length not exceeding 6 feet, and a "tee" handle which is labeled "engine primer". The primer shall be furnished with a temperature control.

3.10.2.2 Alternator. A heavy duty splashproof alternator with double mounting lug shall be provided. The alternator shall be not less than 50 amp capacity and shall provide charge at engine low idle speed. The regulator shall be integral with the alternator.

3.10.2.3 Starting system. The starting system shall be 24 volts and shall include, as a minimum, a starting motor solenoid, a starting motor, necessary wiring and cable, starter relay and starting motor switch. A means shall be furnished to prevent energizing the starter motor whenever the engine is running. The crane shall be provided with either an interlock in the starting system or other means to prevent energizing the starter motor except when the transmission is disengaged. The starter shall be in accordance with the engine manufacturer's recommendations. The starting motor and starter solenoid shall be splashproof.

3.10.2.4 Fuel system. The capacity of the fuel tank shall provide for not less than 10 hours operation when tested as specified in 4.6.2.30 and 4.6.2.29. The fuel-tank filler cap shall be captive chained or hinged. The fuel fill port shall be labeled "Diesel Fuel" with cast or embossed letters in not less than 1-inch letter height. The tank shall accept fuel at the rate of not less than 24 gallons per minute and shall be located so that the tank can be filled from a 5-gallon can conforming to MIL-C-1283 by a person standing on the ground or a surface on the crane intended for standing. The tank shall be able to accept a fuel nozzle of 2 inches diameter with sufficient air release clearance. The fuel tank shall be equipped with a drainplug at the lowest point of the tank. The drainplug shall

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be removable with handtools without the removal of any other component and shall be located such that the tank contents can be drained into a container. The fuel tank shall be protected by position or otherwise from external damage. Fuel lines shall provide continuous operation at full throttle and maximum loads, including slope ascension. The fuel system shall include a fuel transfer pump, fuel strainer, water separator with provisions for draining sediment, fuel filter(s) in accordance with engine manufacturer's recommendations and a fuel injection pump, including provisions for priming. The fuel strainer or fuel filter(s) shall be between the fuel tank and fuel transfer pump. A fuel shutoff valve in the fuel line shall be provided between the tank and the fuel filter(s) if fuel filter(s) are located below the top of the fuel tank.

3.10.2.5 Air cleaner. The engine shall be furnished with a heavy-duty, dry-type air cleaner in accordance with MIL-A-52363, class II or class III. The cleaner shall be of the dual-element type and shall include a dust scavenger. The capacity of the air cleaner shall be in accordance with the engine manufacturer's recommendations. An air-cleaner intake-restriction indicator shall be provided and shall be visible from the operator's compartment. The intake-restriction indicator shall have graduated markings with continuous reading on remaining service of the air cleaner with and without the engine running. The indicator shall also have reset capabilities.

3.10.2.6 Cooling system. When a liquid-cooled engine is furnished, the cooling system shall be the closed-pressure-type, incorporating a radiator with a radiator pressure cap, hoses, thermostat, fan, and circulating pump. Under all operating conditions specified herein, the temperature differential between the cooling system (radiator top tank) and ambient air temperature shall be less than the difference between the boiling point of the pressurized system and 120° F. Drains shall be provided to completely empty the cooling system. Petcocks on all drains shall be accessible and contents should drain into a container. The cooling-system fluid shall be a solution of ethyleneglycol and water conforming to MIL-A-46153. All fill ports shall be accessible. When an air-cooled engine is furnished, the engine shall be cooled by a fan that is either gear-driven, belt-driven, or directly driven from the crankshaft. The air intake face of the fan shroud shall be provided with a removable grille or screen guard.

3.10.2.6.1 Fan. The fan shall either discharge away from the operator, or be a suction type used in conjunction with a radial flow type shrouding such that the air drawn through the radiator is redirected in a radial direction around and about the radial shrouding and then expelled out the sides of the engine compartment. The fan shall rotate on antifriction bearings. When utilized, the fan belt takeup shall be at least 3 percent of the belt circumferential length, plus an allowance for replacement of new belts without forcing. The belts shall be clear of all other components within the adjustment range. Belts shall be accessible for easy replacement. If the fan operates below the 30 inch water fording, provisions shall be made for disconnecting the fan when necessary for water fording

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operations without affecting coolant circulation within the engine. Fan disengagement, shall be controlled from the operator's compartment with a control labeled "fan disengagement". A welded wire or expanded metal fan guard shall be provided.

3.10.2.6.2 Water pump. When utilized, the water pump(s) shall be of the centrifugal type, shall be replaceable, and shall be equipped with shaft seals that will prevent leakage. The shaft shall be corrosion resistant when exposed to water and a mixture of ethylene-glycol and water. Either a lifetime lubrication system or accessible pressure lubrication fittings shall be provided.

3.10.2.6.3 Thermostat. When utilized, the thermostat shall be in accordance with the engine manufacturer's recommendations. The thermostat shall be removable.

3.10.2.6.4. Radiator. When utilized, the radiator assembly shall be a heavy duty type and shall consist of a core with an expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator cap shall meet the requirements of SAE J164 and be accessible without removal of the engine hood. A sign warning "CAUTION - DO NOT OPEN WHEN HOT" shall be placed near radiator cap. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced in normal operation of the crane. The radiator shall be protected by a heavy grilled guard or by the counterweight. A shield shall be provided, if necessary to protect the radiator from mud and gravel thrown by the tires. Hoses shall clear the crane structure and other components under all operating conditions specified herein. The cooling system shall include baffles and shroud to prevent recirculation of cooling air.

3.10.2.6.5 Gaskets, seals, and packing. Gaskets, seals, and packing used in the cooling system shall be made of materials capable of resisting deterioration from mixtures of water and ethylene-glycol-base antifreeze conforming to MIL-A-46153.

3.10.2.6.6 Engine coolant hose. All engine coolant hoses shall be SAE J20, heavy-duty type. All hose clamps shall be stainless steel type in accordance with SAE J536, type F, style 1, 2, or 4.

3.10.2.7 Governor. A governor shall be provided to limit the engine speed to not more than the intermittent duty speed recommended by the engine manufacturer when the crane is operated under any condition specified herein.

3.10.2.8 Exhaust system. The exhaust system shall be protected against entry of rain. The back pressure of the exhaust system, including muffler, shall in no case exceed the maximum recommended by the engine manufacturer when measured approximately 2 inches beyond the header outlet at any load up to rated net continuous load. The exhaust system shall be arranged or shielded to prevent burns to the operator and maintenance personnel while working on the step surfaces of the crane. Exhaust discharges shall be directed so as to minimize the exhaust fumes at the operator's station. The exhaust system shall

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not obstruct the view of the operator. The exhaust system shall be provided with a spark arrestor conforming to Forest Service Standard 5100-1 and SAE J350 for not less than 80 percent efficiency. The spark arrestor shall be provided with a means for cleaning out trapped particles.

3.10.2.9 Crankcase ventilation system. The crankcase ventilation system shall prevent water entering the engine during fording operations. When the crankcase vapors are directed into the induction system, means shall be provided on turbocharged engines to prevent coking of turbocharger blades. Vapors or condensation shall not flow through the air-cleaner element.

3.10.2.10 Oil filter. Oil filter(s) of the heavy-duty full flow type or heavy-duty full-flow and partial flow type shall be furnished in accordance with the engine manufacturer's recommendations. A label specifying filter type shall be placed near the point of filter installation. Oi1 lines, when applicable, to and from the filters shall be installed to minimize vibration and shall be of flexible hose meeting the requirements of SAE J30(d) with reusable fittings.

3.10.2.11 Tachometer drive. A heavy-duty key type tachometer drive shall be provided and shall have a 5/32 inch female connection in accordance with SAE J678 for diagnostic purposes. The speed of the tachometer drive shall be 1/2 of the crankshaft speed.

3 10.2.12 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained metal threaded cap which provides a positive seal of the sampling port. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. The sampling valve shall be labeled adjacent to the valve indicating the type of oil sampled (engine or transmission, whichever is appropriate).

3.11 Drive assembly. The drivetrain assembly shall consist o£ all components necessary to transmit power from the engine to all four wheels. Rear or front wheel drive disconnect may be provided forhigh speed operation. All rotating shafts and axles shall be supported on antifriction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system shall be provided for other friction parts. Gear housing shall be equipped with filler plugs and magnetic drainplugs. Differential ring and pinion gears shall be adjustable. Drive shafts shall be provided with a minimum of two heavy-duty universal joints and a nonbinding slip joint. Drivetrain components shall conform to and shall be installed in accordance with each component manufacturer's recommendations. Drivetrain universal Joints shall not exceed 180°F when the crane is

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tested as specified herein. The axle manufacturer's rating for torque input, gearing and beam loading shall be equal to or exceed the actual maximum loads imposed under all operating conditions.

3.11.1 Transmission. The crane shall be provided with a hydrodynamic power shift transmission as defined in SAE J645. The crane shall be capable of accelerating from a standstill to maximum speed under continuous full engine throttle on level ground and this shall be accomplished without damaging the transmission. The transmission shall be protected from damage when the operator moves or attempts to move the directional control to the opposite direction while the crane is in motion. The transmission oil shall conform to MIL-L-2104, Grade 10, or MIL-L-46167. The fill tube shall be functionally labeled. All transmission component material shall be compatible with these oils. The transmission fluid shall be filtered by filter(s) having replaceable elements in accordance with the transmission manufacturer's recommendations. The transmission controls shall include a disconnect valve.

3.11.1.1 Hydrodynamic powershift transmission. The transmission shall provide controlled application of the clutches to provide smooth shifting, limiting slippage which would damage the clutches. All transmission gears shall be constant mesh. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a mechanically driven pump. The transmission shall be capable of withstanding maximum stall operation for a minimum of 30 seconds without damage or permanent deformation and without exceeding the fluid temperature limit of 250° F with an initial fluid temperature of not less than 157° F in an ambient temperature of not lower than 25° F.

3.11.2 Rear axle oscillation. The rear axle shall oscillate freely such that the total vertical travel of each rear wheel shall be a minimum of 12 inches (six inches above and six inches below the horizontal axle centerline.)

3.12 Hydraulic system. The hydraulic system shall consist of all fluidpower components necessary for the operation of the crane, including the steering system but does not include the brakes or the hydrodynamic type power train. The hydraulic system shall provide for a simultaneous swing and load hoist. All filler tubes shall be labeled. The working hydraulic system (boom lift, winches, boom extension, slewing and steering) maximum operating pressure (relief valve setting) shall not exceed 3000 psi. The system fluid shall conform to MIL-L-2104, Grade 10 or MIL-L-46167. All hydraulic system component material shall be compatible with these fluids. The temperature of the hydraulic fluid shall not exceed 100° F above the ambient during all operations and tests specified herein. Hydraulic fittings on pumps, valves and cylinders shall be either the O-ring, straight thread type or 37 degree flare swivel type, as specified in SAE J514 or the 4-bolt flange type as specified in SAE J518. When 37 degree flare fittings are used on a hydraulic line, the mating connection on the components shall also be a 37 degree flare type. The hydraulic system shall be flushed free of foreign matter including weld slag and spatter. The connection ports of the

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hydraulic components shall be sealed by the component manufacturer. In event of failure or damage to the hydraulic system, a means to prevent the boom from automatically lowering or retracting, and the load from lowering shall be provided. Means or techniques shall be provided to allow for
manual override for retracting and lowering the boom and load not in excess of 10 ft/min and swinging the superstructure in the event of hydraulic power loss or line failure. A straight thread "O" ring boss diagnostic port conforming to SAE J514 shall be provided at the pump(s) outlet or pressure inlet of each control valve. The hydraulic system shall provide for fluid passage between the carrier and the superstructure with continuous 360-degree superstructure rotation. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the crane and shall be protected by structural members from striking obstacles.

3.12.1 Hydraulic lines. Hydraulic lines shall consist of tubing, flexible hose and steel fittings. Hose or a combination of hose and tubing shall be used between components that are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance and maximum protection. Provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines, other than suction lines, and cylinder bleeder lines, shall be considered pressure lines. All lines shall be flushed free of foreign matter.

3.12.1.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 or MIL-H-52471. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. The crane shall be compatible with hose assemblies with field attachable reusable fittings in place of each hose assembly with permanently attached couplings, when tested as specified herein. All fittings shall be 37-degree flare, female swivel or 4-bolt split flange as specified in SAE J516. Hose ID shall be limited to 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. The hose selected shall have a rated operating pressure that is greater than the hydraulic circuit maximum relief valve setting. The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.12.1.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525 and shall meet the requirements specified herein. Pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inches. Tubing shall have a minimum burst pressure that is equal to or greater than 3.0 times the system maximum working pressure. Tubing shall be securely clamped. All bends shall be smooth without flattening, kinking or wrinkling of the tube. The term "without flattening" is defined as follows:

> Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum

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diameter minus the tube minimum diameter shall be not greater than 15 percent of the nominal tube diameter.

3.12 1.3 Pressure tube fittings. Unless otherwise specified herein pressure tube fittings shall be 37-degree flared and straight thread fittings conforming to SAE J514 or 4-bolt split-flange type conforming to SAE J518. Jump size tees may be used. The 4-bolt split-flange type connections may terminate in either a flange head that incorporates a SAE J518 plain mating face or a flange head that integrally incorporates four boltholes and an O-ring groove for mating to a SAE J518 port face. Pressure tube fittings that incorporate the SAE J518 connection shall contain a boss into which the tube shall be brazed or welded. Pressure tube fitting sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, and 2 inch. Pressure tube fittings shall have a minimum burst pressure 3 times the maximum operating pressure.

3.12.1.4 Suction lines. A pump inlet suction line shall include provision for flexibility when there is vibration or movement between the reservoir and the pump. The suction line shall be sized so that the pump inlet pressure does not drop below atmosphere by more than 5 inches of mercury vacuum at 140° F ±10° F oil temperature and at governed engine speed. The inlet suction line installation shall include no stress concentration at the end fittings and no kinks. Hose shall conform to SAE J517, SAE 100R4 or shall be oil resistant hose. Suction hose clamps shall be stainless steel type in accordance with SAE J536, Type F, Style 4 or saddle worm thread type with stainless steel inner liner or a heavy duty T-bolt clamp, or shall be in accordance with fittings specified in 3.12.1.1.

3.12.1.5 Return knee. Return lines shall be in accordance with SAE J517.

3.12.2 Hydraulic pumps. The hydraulic pump(s) shall have been tested in accordance with NFPA Recommended Standard T.3.9.18-1976. When tested in this manner, the contaminant tolerance profile for the hydraulic pump(s) shall either:

a. Not fall below or cross the B_{10} curve of the hydraulic system filter when tested in accordance with MIL-F-52723 (see 4.6.2.17.2).

or else

b. The pump Omega rating shall be less than the filter Beta 10 rating.

3.12.3 System filtration. System filter(s) shall be furnished either in the pressure line, return line, or combinations (as specified in 3.12.3.1 or 3.12.3.2). Labels identifying each filter shall be provided near the points of filter installation.

3.12.3.1 Return line filters. Although the system filter element need not be qualified to MIL-F-52723, it shall meet the performance

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requirements as stated therein. The filter shall maintain a full flow condition for the duration of reliability testing as specified in 4.6.2.30. The filter bypass may open for system oil at a temperature less than 70° F. The filter element shall have been tested to the performance requirements of MIL-F-52723 at a flow rate equal to or greater than the crane maximum system oil flow at maximum governed engine speed. All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed, without removal of other components, for examination or replacement without spilling or draining oil from the reservoir. At the maximum system oil flow through the filter, all valves in the neutral position, at maximum governed engine speed and an oil temperature of 150° F±5° F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the system filter and shall open at not less than 10 psi greater than thc pressure drop measured across the new filter assembly and it shall permit full flow at a pressure drop not greater than 35 psi. The filter housing shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage. The system filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the filter clement is clean and when the system filter is beginning to bypass, requiring a change of the element.

3.12.3.2 Pressure filter. The pressure filter(s) shall conform to 3.12.3.1 except that proof pressure test shall be 150 percent of the maximum system operating pressure.

3.12.4 Reservoir. A hydraulic reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall includea filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The Add and Full marks on the dipstick or sightglass(s) shall be in accordance with SAE J614b and indicate the correct oil level when the crane is on level ground, and the oil is at normal operating temperature. The fillercap and tube shall be located to provide for filling from a standard 5 gallon container conforming to MIL-C-1283 by a person standing on the ground or a surface on the crane intended for standing, and to prevent entrance of contaminants throughout filling and checking operations. The reservoir filler neck shall contain a removable 50 mesh, or finer, strainer retained in place. The filler tube labeled "hydraulic fill" shall be not less than 1-inch inside diameter. The reservoir pump inlet line suction openings except for the auxiliary steering pump opening, shall be located a minimum of 1.25 inches above the bottom

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of the reservoir. When furnished, the auxiliary steering pump inlet line suction opening shall be located a minimum of 1/2 inches above the bottom of the reservoir. Except for sealed reservoir, a pressurized breather with a 30 psi maximum relief valve and 1 inch Hg vacuum breaker shall be furnished. The breather shall have a nominal rating of 40 microns. The breather shall preclude the entrance of water. The reservoir shall provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow except the vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir. When filters are located within the reservoir, a removable 50 mesh, in-tank suction strainer with bypass valve shall be installed in the suction line of the reservoir. The reservoir shall have suction flanges and one or more access holes with covers for inspection, cleaning and maintenance of the entire inside.

3.12.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions.

3.12.6 Hydraulic control valves. The control valve spools shall be spring centered with sufficient force to return and retain the levers at the center position throughout all crane operations. The spool connection eye end shall conform to SAE J748 and shall not bind while operating as specified herein. The exposed control-valve spool surfaces shall show no evidence of corrosion or pitting when tested as specified herein. The contractor shall present information during first article test that the control valves were tested in accordance with procedures specified in SAE J747 and J1117. The inlet to outlet port pressure differential shall not exceed 3 percent of the maximum operating pressure at rated flow.

3.12.7 Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seamless steel tubing or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder component. All hydraulic system and steering cylinders shall perform as specified herein without buckling or bending. Each piston shall have a bearing surface concentric with the piston to carry the piston side loads against thc cylinder walls. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder. All cylinder rods shall be hard chrome plated. All cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, internal or external leakage.

3.12.8 Contamination levels for hydraulic system. The particulate matter in hydraulic fluid removed from the system directly upstream of the filter, shall not exceed the ISO code level of 19/16 in

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accordance with SAE J1165 after completion of assembly of the crane when tested as specified in 4.6.2.17.3.

3.12.9 External leakage. After completing tests specified in 4.6.2 the degree of external hydraulic system leakage shall not exceed a class 4, for dust-free conditions and a class 4D for dusty conditions of SAi Jl176. External hydraulic system leakage shall not exceed a class 3 for production cranes under normal operation.

3.13 Superstructure. The revolving superstructure assembly shall be capable of continuous 360-degree operation. The superstructure shall be capable of a swing speed of not less than 1.75 rpm and not more than 3.5 rpm at governed engine speed with the swing control on the full open position and shall be equipped with a swing brake and lock. The counterweight shall be made of a homogeneous material.

13.13.1 Pedestal. The pedestal and base shall be mounted on the chassis to insure continuous 360-degree working range and shall withstand the forces caused by lifting, swinging with rated loads, and traveling without loads.

3.13.2 Boom assembly. The boom shall be of the hydraulically operated telescoping type, consisting of the base section and one or more sliding sections to reach the operating radii specified in 3.9.9. Extension and retraction of the boom sections shall be accomplished by one or more hydraulic cylinders. If cables or chains are used in the boom assembly, data shall be provided by the contractor illustrating the capability of the cables and chains to withstand a force of 3.5 times the total force in the cable chain system when extending the boom at maximum lift angle carrying rated load. The crane shall incorporate means to prevent accident or damage from two-blocking. The prevention of two-blocking shall be accomplished by de-energizing the boom extension and load hoisting functions. The boom lift shall have combination controls for hand and foot operation.

3.13.3 Hook block. The crane shall be provided with a main and an auxiliary hook block. The hook blocks shall be of the lubricated type with swivel hook and safety latch. The main hook block shall be rated not less than 40 tons and the auxiliary hook block shall be rated not less than 8 tons. Both hook blocks shall be provided with a bucket for ending the hoist cable at the hook block. The sheave shall conform to SAE J881. All sheaves carrying hoist lines shall be guarded to prevent the rope from leaving the sheave groove during operation. The hook blocks shall have adequate weight to insure payout of the wire rope without load on the hook when the boom and hook block are in any position. The contractor shall provide certification of the hook blocks' capacities at the time of first article test.

3.13.4 Winch. The crane shall be provided with a hydraulically driven main hoist drum and an auxiliary hoist drum. The assemblies shall include a self-locking mechanism or self-energizing brake capable of holding their respective maximum rated loads. The drums

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pitch diameter shall be not less than recommended by SAE J881 for the wire rope furnished with the winches.

3.13.4.1 Powered load lowering. The main and the auxiliary load hoist drums shall be equipped with a powered reversing drive, and suitable control features to provide precision controlled power lowering of loads. The powered load lowering mechanism shall be capable of lowering the loads at a smooth, controlled rate of speed, dependent on engine speed, without jerking, bouncing or slipping the load. A braking device may be used to assist the power lowering mechanism. The winch shall not be capable of free fall.

3.13.5 Wire rope. The crane shall be provided with sufficient rope for both the main and auxiliary systems so that when the boom is fully extended at its maximum boom angle, reeved for maximum part lines and the hooks at ground elevation, there shall be not less than one complete layer of rope on the hoist drum. The wire ropes shall be in accordance with RR-W-410, type I, class 2, 6 x 19, construction 3 or 4, IWRC improved plow steel uncoated, preformed, regular lay. The wire ropes shall have safety factors not less than recommended by SAE J959.

3.13.6 Boom angle and boom length indicators. A boom angle indicator shall be provided which shows angular elevation of the boom and conforms to SAE J375. The boom angle indicator shall be visible to the operator in his normal seated position. A boom length indicator shall also be provided in increments to match with the load capacity chart in accordance with SAE J1180.

3.13.7 Load indicating and warning device. A load indicating and warning device of the load-moment type shall be provided to perform the following functions while on outriggers:

a. Provide a dial or digital reading of hook load in pounds.
b. Provide a dial or digital reading of load rating in feet.
c. Provide a dial or digital reading of load moment as a percentage of rated capacity.
d. Provide visual warning (amber light) when load-moment reaches 85 percent of rated capacity, visual and audible (red light and alarm) warning at a 95 percent rated capacity, and control lock out for boom lowering and extension at 100 percent rated capacity.

The device shall provide for manual selection of parts of line. The device shall conform to SAE J159, J375, and J376.

3.13.8 Lift and extension mechanism. The crane shall be capable of accomplishing hoist and boom extension both simultaneously and independently. Control valves or other positive means shall be provided to prevent cavitation in the lifting/lowering cylinders and the extension/retraction cylinder(s) when tested in accordance with 4.6.2.4 and 4.6.2.38.

3.14 Outriggers. Four independently hydraulically operated outriggers, capable of extension to not less than 7 inches below

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ground level, shall be provided. The outriggers shall be operated from the operator's compartment and shall not collapse due to loss of hydraulic pressure in the supply lines. The outriggers floats shall be attached to the outriggers beams by swivel joints which permit displacement of the floats by not less than 3 inches between opposite sides on opposite corners (if applicable) of the floats without breakage or deformation. An indicator shall be provided for the operator to determine levelness of the crane during operation of the outriggers. Means shall be provided to prevent the operator from accidentally retracting the outriggers. The outrigger floats shall be sized such that the load bearing surface area per float is not less than 660 square inches.

3.15 Wheels and tires.

3.15.1 Wheels. The crane shall be mounted on four hubs with four demountable rim and tire assemblies as recommended by SAE J751. The wheels shall be fastened to the hubs in accordance with the axle manufacturer's recommendations for off highway application. Wheel rims shall be in accordance with the TRA Yearbook.

3.15.2 Tires. All tires shall be new and unused, wide base off-the-road type, and of identical size and construction. The tires shall be pneumatic, low pressure, tubeless, furnished with type E-3 tread. Tires shall have a section width of not less than 29.50 inches and a ply rating of not less than 26. Tire loadings shall be in accordance with the TRA Yearbook and the tire manufacturer's published literature. Valves and caps shall be in accordance with the TRA Yearbook. Large bore valves shall be furnished. A valve adapter shall be furnished to reduce the large bore to the standard passenger vehicle size bore. Valve cape stroll be metal with elastomer inner seal. Hydroinflated tires are not acceptable.

3.16 Wheel guards. Wheelguards shall be furnished at all four wheels to provide protection for the operator from material being thrown radially off the rotating tires. Each fender shall structurally support a weight of 300 pounds without evidence of damage.

3.17 Steering and steering control. The steering method shall be the Ackermann type steering and shall provide the operator with a choice of frontwheel, or a combination of front- and rear-wheel steering, including crab steer capabilities. An indicating light shall be provided on the instrument panel to indicate when the rear wheels are positioned other than straight ahead. The steering shall be accomplished by fluid power actuators. The steering system shall permit control of the crane while traveling at any speed up to the maximum attainable travel speed (see 3.9.3). The power steering actuating device shall be located so as to absorb all road shock without transmitting the shock effect to the steering wheel. The crane shall operate without loss of operator control when tested as specified in 4.6.2.11.

3.17.1 Steering control. The front wheel steering system shall be controlled by a single steering wheel. Rear wheel steering shall be

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controlled by the same steering wheel or a tiller. The effort required to turn the steering wheel shall not exceed 10 pounds pull at the rim. The steering wheel shall require not more than 6 turns to move the wheel from one stop to the opposite stop. In the two wheel steering mode the steering wheel shall require not more than 10 seconds for the maximum range of movement at low idle engine speed and shall require no less than 5 seconds at governed engine speed. There shall be a minimum of 7 inches clearance measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). The steering wheel shall be capable of withstanding a minimum 300 pound force applied perpendicular to the plane of the steering wheel, and a minimum 150 pound force applied tangentially to the rim of the steering wheel, without permanent deformation or damage.

3.17.2 Emergency steering. Emergency steering conforming to SAE J53 shall be provided for retaining steering control in the event of engine failure or steering power source failure when traveling at any speed to the maximum speed.

3.18 Brake system. Service and parking brakes shall be provided and shall be independently actuated systems. If a separate hydraulic system for the brakes is utilized, the brake system shall be filled with a fluid that is listed on the qualified product list of MIL-B-46176. If air brakes are provided, air brake service and emergency line coupling conforming to SAE J318 shall be provided on the rear of the vehicle, a relay emergency valve and an air dryer shall also be provided.

3.18.1 Service braking system. Unless otherwise specified herein, controlled service brakes shall be provided and shall conform to SAE J1152 for cranes. The brakes shall be capable of bringing the crane with no load to a complete stop without any wheel leaving the ground from maximum crane speed as follows:

a. Dry - within distances specified in table 7 of SAE J1152 with fishtailing of no more than 1 foot when tested as specified in 4.6.2.14.
b. Fade - within distances specified in table 7 of SAE J1152 with fishtailing of no more than 1 foot when tested as specified in 4.6.2.14.
c. Wet - Within 10 feet greater distances than those specified in table 7 of SAE J1152 with fishtailing of no more than 1 foot when tested as specified in 4.6.2.26.

Pedal force while making any stop shall not exceed 90 pounds. The brake system shall provide for bleeding of air from hydraulic lines and for draining of moisture from the compressed air system, as applicable. Automatic or manual brake adjustment provisions shall be provided. Automatic adjustment shall maintain correct lining to drum or disk adjustment without looseness or lockup. Automatic adjusters shall be enclosed or guarded for protection from environmental damage or malfunction. The brake system shall withstand an operator's brake

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application force of not less than 300 pounds a minimum of 10 times, without permanent deformation or evidence of mechanical damage.

3.18.2 Parking system. The parking brakes shall provide on-grade parking performance, but not necessarily stop the gross weight of the crane without load when the crane is stopped on a 25 percent grade in both the ascending and descending attitude when tested in accordance with test method 6. The parking brakes shall be hand- or foot-operated and equipped to remain applied until manually released by the operator. The manual control shall be located within reach of the operator and in a noninterfering position. Hand-lever pull to set the parking brake shall not exceed 50 pounds. The foot pedal force shall not exceed 75 pounds. The parking brake shall be an independent mechanical friction brake mounted on the drive or propeller shaft rotating with the wheels, or a separate isolated system actuating the service brakes. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any component parts.

3.18.3 Emergency braking provisions. The crane shall provide not less than six "dead engine" stops while descending a 25 percent grade. The emergency braking provisions may be provided through either the parking brake or Service brake system. Means shall be provided to release the brakes for towing.

3.19 Operator access. Operator-access systems to the crane shall be in accordance with minimum requirements of SAE J185.

3.20 Operator's seat. A cushioned bucket type operator seat shall be furnished in accordance with minimum requirements of SAE J899 and located in accordance with SAE J1163 to permit 90 percent of US male/female operator population (SAE J833) to reach controls in the optimum areas of the crane as defined in SAE J898.

3.21 Cowl or hood. The engine shall be protected by a hood cover and sideplates, except that sideplates may be omitted when the wheel guards provide the engine compartment with protection from road splash and debris. The sideplates shall be either the quick opening or the bolted type with quick access openings for the performance of daily maintenance checks. The sideplate shall be capable of being opened or removed by one person wearing gloves. Elastomer hood latches or paddle latches shall be provided on the quick opening type sideplates or the quick access opening of the bolted type sideplates. The elastomer latches shall withstand an air ozone rating of two or better in accordance with ASTM D1171. Handtools shall not be required to release these latches. All engine accessories shall be completely accessible without requiring the removal of any additional crane components when the hood cover and sideplates are opened/removed.

3.22 Instrument panel. The instrument panel shall be located inside the operator's compartment so that the operator can read the instruments from his normal operation position. The instrument panel shall be installed on an incline at a minimum of 5 degrees to a maximum of 80 degrees. On this panel shall be mounted the charging system monitor, engine oil-pressure-monitor, engine-coolant temperature monitor, rear steer indicator warning light, torque

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converter or transmission-fluid temperature monitor, fuel gage, and any other instruments used, unless otherwise specified herein. Universal symbols shall be furnished in accordance with SAE J298 and ANSI MH11.3. All gages shall be flush mounted and shall be provided with switch-controlled illumination.

3.23 Controls and instrumentation. Each control shall be provided with symbolic labels either on plates conforming to 3.32 or embossed or depressed on or near the control. Labeling for levers which control any part of the lift mechanism shall conform to ANSI MH11.3. The labels shall be visible to the operator from his normal operating position. All controls for the crane shall be located in the operator's compartment and shall be within the maximum area specified in SAE J898. Electric controls and instrumentation shall be weathertight. All hand operated control shall be operable by persons wearing cold-wet gloves. Mechanical push-pull remote controls, when furnished, shall be of waterproof, heat resistance, antifriction type, conforming to requirements for marine cables specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cables shall be fabricated of corrosion-resistant steel that has been passivated as per QQ-P-35. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. The "as-installed" mechanical efficiency of each installation shall be not less than 70 percent and the no-load friction shall not exceed 5 ounces per foot. The control manufacturer's recommended load ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Only controls associated with the cab, such as windshield wiper and heater controls, may be mounted on the cab structure.

3.23.1 Charging system monitor. A remote indicating ammeter or voltmeter, or alternator indicator red light shall be furnished. The ammeter shall be of sufficient capacity to indicate variations in the current supply or demand experienced by the crane, exclusive of starter requirements. The indicator light shall light when the alternator is not charging and the engine run control is "on" and shall be labeled "alternator" or with appropriate universal symbology.

3.23.2 Engine oil pressure monitor. A remote indicating engine oil pressure gage shall be furnished. A low pressure red indicator light or low pressure alarm shall be furnished in addition to the gage. Means shall be provided to check operation of the light or alarm.

3.23.3 Engine coolant temperature monitor. For liquid cooled engines, a remote indicating engine coolant temperature gage shall be furnished and a high temperature red indicator light or alarm shall be furnished in addition to the gage. Means shall be provided to check operation of the light or alarm.

3.23.4 Rear steer indicator. An indicating light shall be furnished to indicate when the rear wheels arc positioned other than straight

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ahead. Means shall be provided to check operation of the light.

3.23.5 Transmission temperature monitor. A remote indicating torque converter or transmission temperature gage shall be furnished. Ahigh temperature red indicator light shall be furnished in addition to the gage. Means shall be provided to check operation of the light.

3.23.6 Air pressure monitor. When a compressed air system is furnished, an air pressure gage shall be furnished and an audible alarm which exceeds operating noise levels by 20 dB shall indicate when air pressure is inadequate for proper brake operations.

3.23.7 Hourmeter. An hourmeter which registers the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hourmeter shall be accessible for reading without removing any component of the crane.

3.23.8 Fuel monitor. A fuel gage shall be provided.

3.23.9 Engine run control. An engine run control stroll be furnished and shall be a keyless type. It shall have not less than two positions: "On" and "Off". The switch shall control the actuation of all electrical monitors and electrical equipment except the horn.

3.23.10 Light switches. "On-off" light switches shall be furnished on the instrument panel. One switch shall control the forward flood lamps, one switch shall control the rear floodlamps and one switch shall control the boom floodlamps(s) as specified in 3.24.4. All internal and external lights, the horn, and the backup alarm shall be made inoperable when blackout lights are used (see 3.24.4.1) except for warning lights, such as temperature and oil lights, which shall remain operative in the blackout mode. One light switch conforming to MS51113 shall be provided for the headlamps (see 3.24.4), blackout headlight (see 3.24.4.1), two blackout stoplight-taillights (see 3.24.4.1), and instrument panel lights (see 3.22).

3.23.11 Braking pedal. The brake pedal shall be located for convenient right foot operation and shall provide braking control only. The pedal shall be a minimum of 2-1/2 inches wide and shall have an all around clearance of not less than 2 inches except that 1 inch minimum clearance is acceptable next to the steering column providing the pedal width is a minimum of 3-1/2 inches wide.

3.23.12 Engine speed control. Hand and foot engine speed control throttles shall be provided. The engine speed control pedal shall be installed for comfortable right foot operation and shall be located to the right of and shall be free of interference from the brake pedals. The engine speed control pedal shall have an all around clearance of not less than 3 inches.

3.23.13 Transmission controls. Selective forward and reverse and speed range controls shall be provided and shall be located for convenient hand operation. Forward, reverse, and speed ranges shall be labeled in accordance with SAE J298. Movement of the control lever in a forward or upward direction shall cause forward motion of the

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crane. Transmission control levers and linkage shall withstand, without permanent deformation or damage, a force of 100 pounds at the handgrip and the shock and vibration encountered during operation specified herein. The force necessary for the movement of the hand control levers shall not exceed 20 pounds.

3.23.14 Hydraulic controls. Boom lift/lower and load hoist/lower shall be located for convenient right hand operation. All other hydraulic controls shall be located in accordance with ANSI B30.5 All hydraulic controls shall be of the self-centering lever(s) type. Control labeling shall be in accordance with SAE J298. All control levers and linkage shall withstand, without permanent deformation or damage, a force of 75 pounds at the handgrip and the shock vibration encountered during operations specified herein. The force necessary to move any control to the end of its travel shall not exceed 20 pounds and shall have a minimum force of 2 pounds at the handgrip. Movement of the control levers shall be as stated in ANSI B30.5.

3.23.15 Starter switch system. The starting system solenoids (see 3.10.2.3) shall be operated by a pushbutton or a spring-loaded starter switch. If a spring-loaded starter switch is provided it may be incorporated as part of the engine run control (see 3.23.9). The starter switch shall be inoperable when the engine run control is in the "off" position.

3.24 Electrical system. The crane electrical system shall be for heavy duty, 12 volt or 24 volt service and in accordance with SAE J539. The electrical circuits shall be assigned identification numbers or be color coded as shown in table I of SAE J821. All electric wiring shall be routed so as to provide ease of maintenance and maximum protection.

3.24.1 Batteries. Batteries shall be furnished and shall be in accordance with MS35000, type 6TN and shall be readily accessible for service. The batteries shall be negative grounded in accordance with SAE J538. The battery terminals shall be accessible for removal without requiring disassembly of other components.

3.24.1.1 Battery mounting. Batteries shall be located so they can be cleaned, serviced, and removed without removing any component except the quick-release battery box cover if a battery box is furnished. Battery supports, holddowns, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with a coating conforming to TT-C-494, type II. The battery shall be mounted in such a manner as not to interfere with access to engine components (accessories). Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall be provided to hold the battery in a fixed position. The battery compartment (if furnished) shall have provision for drainage and provision for gas venting at or near the top of the compartment. Cover and positioning shall be protected against short circuiting. Ungrounded cable shall be protected by rubber grommets or insulated

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passages at entry to the battery box and by nonconductive boots at the terminal end.

3.24.1.2 Battery cable. Battery cables shall be furnished which meet the requirements of SAE J1127. The voltage at the storage battery terminals and the starting motor terminals including connections shall not differ more than those shown in table I of SAE J541. Positive and negative cable terminals shall be identified and corrosion-resistant SAE bolts and nuts provided.

3.24.2 Slaving components. The crane shall be equipped with a 24-volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the crane batteries and slave starting of the engine from an external power source and shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the crane near the battery enclosure and shall be accessible to personnel standing on the ground. A plate shall be furnished adjacent to the slave receptacle which reads "24 Volts". The plate shall be in accordance with 3.32.1.

3.24.3 Circuit breaker. Each electrical circuit shall be protected with a circuit breaker in accordance with SAE J553 and shall have labels which describe the function served by the circuit breaker. Fuses are not acceptable.

3.24.4 Lights. The crane shall be provided with not less than two headlamps for forward illumination and not less than two headlamps for rearward illumination. The lights shall be shock mounted in elastomer ring housings and shall conform to SAE J598. Headlamps shall conform to SAE J1029. Not less than two floodlamps shall be provided; one shall be located on the end of the fixed portion of the boom and directed for ground illumination, and the other shall be adjustable and located BO as to illuminate the hook block in any position Thc adjustable floodlamp, shall be adjustable by the operator from inside the operator's cab. Lights positioned in such a way as to be subject to damage shall be protected by guards.

3.24.4.1 Blackout lighting. A separate wiring harness may be provided for the blackout lights. One blackout headlight conforming to MS51318 shall be mounted on the extreme left within the plan outline of the crane at the front, positioned to provide illumination when the boom is in retracted position. The blackout headlight shall be adjustable in accordance with SAE J598. Two blackout stoplight-taillights conforming to MS51330 or M852125 shall be mounted adjacent to the rear taillights, and shall be mounted in 6-inch-diameter holes or shall be provided with guards. Each light shall be recessed not less than 1/2 inch behind the hole or face of the guard. Two blackout marker lights conforming to MS52126 shall be mounted on the front, one each side of the vertical centerline, of the crane at the same level, and as far apart as practicable.

3.24.4.2 Taillights. Two taillight-stoplight assemblies in accordance with SAE J585 and SAE J586 shall be installed, one on the

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left rear and one on the right rear of the crane, with the lens face recessed not less than 1/2 inch back of a protecting member.

3.24.4.3 Interior lighting. The crane shall be equipped with gauge lighting or indicators which are readily visible to the full range of user personnel and be adequately lighted for night operation. Interior lights, gauges and instruments to include warning lights, shall not emit energy out of the 400-625 nanometer wavelength range. All warning indicators shall be light emitting diodes (LED).

3.25 Horn. The crane shall be equipped with an electric, air, or air-over-electric horn. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions. The horn button may be mounted on the steering wheel or instrument panel. The horn shall be distinguishable from and not less than 10dB(A) above the crane noise level when tested in accordance with 4.6.2.22.2.

3.26 Pintle hook assembly and towing lugs.

3.26.1 Pintle hook assembly. A rear mounted swivel pintle hook assembly conforming to MS51117 shall be provided. The pintle hook assembly shall be located on the vertical center line of the crane with a minimum height of 31 inches and a maximum height of 39 inches. The pintle hook assembly mounting shall be of sufficient strength to withstand, without permanent deformation or damage, the drawbar pull induced when the pintle hook assembly is restraining forward motion at transmission stall.

3.26.2 Towing lugs. Two vehicular towing lugs, conforming to MS500004, for tanks (above 50 tons), shall be furnished and shall provide clearance to connect a towbar conforming to MS500048, heavy duty capacity. The lugs shall be installed between l7 to 21 inches on each side of the crane center line, on the rear of the crane.

3.27 Personnel cab. The crane shall be equipped with an insulated, weathertight personnel cab located on the superstructure and shall be in accordance with ANSI B30.5. If necessary for highway and rail transport (see 3.1.3.1 and 3.1.3.2) the cab shall be removable. Quick disconnects shall be provided for all wires required to be disconnected when the cab is removed. The cab space envelope shall be in accordance with SAE J154. The seated operator clearance dimensions shall be based upon a 95 percentile U.S. male wearing cold weather clothing and safety helmet. The cab shall withstand all operational shock and vibration without deformation or damage under operating conditions specified herein. The cab shall be constructed of metal or fiberglass and shall be equipped with a minimum of one door and another unobstructed emergency exit (either door or window that can be opened) on another side of the cab. The emergency exit shall be labeled. All doors and windows that can be opened shall be capable of being opened and closed by personnel wearing gloves. All doors shall have closing latches, except that cab door(s) need not latch open if the door is equipped with a window that can be latched open to provide adequate operator ventilation, opening stops and hold-open

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catches. The doors shall either slide and latch at each end of travel or shall swing open a minimum of 180 degrees and latch open except that if a split door is furnished only the top half must latch open. All hardware shall be of the heavy-duty truck- or industrial-type, resistant to corrosion. Front and rear window defrosters, operating in conjunction with the heater blower (see 3.27.4), or not less than two (2) minimum 6 inch diameter ventilation fans, or the same performance rotating cage type blowers, shall be furnished. The fans shall be directionally adjustable to direct air flow toward all glazing. Each fan shall have an "on-off" switch. Glare reducing side mirrors shall be provided and installed on each side of the crane to provide the operator with rear view visibility.

3.27.1 Safety glazing. Safety glazing of the windshield, door windows, side and rear window shall be provided. The upright seated operator shall have all-around visibility and visibility of the hook block in all possible hook positions. The roof window shall be glazed and shall be sloped to provide for water drainage. Glazing shall conform to ANSI Z26.1, Type ASl for the windshield and roof window and type ASl or AS2 for other windows. Fixed or hinged glazing shall be sealed in waterproof, stress-absorbent molding and movable windows shall be sealed on extra-heavy-type, pile-lined channels.

3.27.2 Interior. The interior of the cab shall be fitted with insulation, retarding both heat loss and noise intrusion. Heavy-duty floor mats shall be furnished. All cab openings shall be sealed with elastomer grommets, seals, weather stripping, or body compound. The elastomer components shall withstand an ozone rating of two or better in accordance with ASTM Dl171.

3.27.3 Windshield and window wipers. The cab shall be equipped with at least two (2) wiper assemblies to clear the windshield and the roof window in accordance with the performance requirements of SAE Jl98. Wipers shall be capable of being operated at two or more frequencies. The frequencies must be maintained during normal engine speed and workload. Wiper arms shall be of the heavy duty commercial type and wiper element shall withstand an ozone rating of two or better in accordance with ASM Dl171. The wiper arms shall provide tension on the glazed surfaces of at least l ounce per inch of wiper arm length. A windshield washer shall be provided for the front windshield.

3.27.4 Heater. A recirculating-air-type heater shall be furnished in accordance with the performance requirements of SAE Jl129. The heater controls shall be within reach of the operator and shall provide for variable selection of blower speeds and temperature. If a fuel-fired heater is furnished, it shall be in accordance with SAE Jl024.

3.28 Electromagnetic interference. The electromagnetic interference emission characteristics of the crane shall meet the limits specified in MIL-STD-461, part 8, class C1, group II, except that a 20 dB relaxation is permitted in the emission limits. The operators horn is exempt from electromagnetic interference requirements.

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3.29 Lubricants. Unless otherwise specified herein, the crane shall be serviced with military lubricants selected by the contractor in accordance with table I, SAE J754. Run-in and testing shall be performed with these lubricants as applicable for the ambient operating temperature. A lubrication data plate identifying military lubricants and applicable temperature ranges shall be provided and mounted (see 3.32.1) on each crane.

3.29.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with handoperated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.29.2 Enclosures with integral reservoir of lubricant. Enclosures such as the gearcases, transmission housing and engine crankcase, which contain a reservoir of liquid lubricants for the lubrication of the parts enclosed, shall be equipped with dipsticks, marked in accordance with SAE J614, check plugs not less than 1/2-inch pipe size, or sight gages to determine the acceptance level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain, except planetaries, shall be fitted with a SAE J531 or J532 magnetic drainplug. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage, except from planetary hubs, shall be to a container on the ground when the crane is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the container. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be provided without the removal or adjustment of accessories or parts. Plates equipped with hand-operated fasteners may be removed.

3.30 Fungus and moisture resistance. Electrical circuitry, including all components and connections except as specified below, shall be protected from the effect of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition specified in 3.30.1:

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components of circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

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3.30.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.31 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions that enable the crane and all removed components to be lifted, without the use of spreader bars, in its transport and operational configuration and to be fastened to the floor or deck of a transportation medium shall be provided. Provisions shall conform to MIL-STD-209, classes 1 and 2 or class 3 for type II equipment.

3.32 Identification marking. Each individual crane and each separable attachment removable for surface transportable purposes (e.g., boom, wheels, cab, winches, hook blocks, or counterweights), shall be identified in accordance with MIL-STD-130 on identification plates conforming to MIL-P-514, type I, style 1, composition C (GG-P-455, type I, grade A, class 1) material. Additional or specific information to be indented or embossed on the plate(s) shall be as specified (see 6.2). Plates shall be furnished and mounted by the contractor. A shipping data plate shall be furnished and shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material and, in addition, shall indicate the silhouette of the crane in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.32.1 Instruction plates/labels. The crane shall be equipped with instruction plates/labels conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material, including warnings, cautions and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the crane. The crane shall also be labeled on or near the data plate "CARC month/year" with the month and year the crane was treated specified.

3.33 Treatment and Painting. The portions of the crane normally painted, both internally and externally, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals and fiberglass, top coat camouflage green 383. Primer in accordance with MIL-P-53030 may be used in 1ieu of MIL-P-52192. Finish coat in accordance with MIL-C-53039 (polyurethane) may be used in lieu of MIL-C-46168. Vendor parts and components received by the prime contractor with either alkyd primer or alkyd paint coating may be coated with above specified primers and paints applied directly over existing alkyd primer or paint without additional service preparation except for cleaning, if the base alkyd primer or paint meets applicable military federal specifications and if the base alkyd primer or paint is allowed to cure for a period of 14 days. This exception for vendor parts and components is not

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intended to supersede the requirement that the complete truck be primed and painted in accordance with MIL-T-704 using the primers and paints specified above. If another color is specified (see 6.2), the above system shall be overcoated with MIL-C-46168 or MIL-C-53039 of the desired color. All surfaces of the crane which exceed 450° F during operation shall not be primed or painted as specified above, but shall be painted with paint conforming to either MIL-P-14105 for steel surfaces or DoD-P-24555 for aluminum surfaces.

3.34 Markings. Registration and other markings shall conform to and shall be applied in accordance with MIL-STD-642. The gross weight of the crane shall be stenciled on each side of the crane. The prescribed tire pressure shall be stenciled in a suitable location adjacent to each tire with block or stenciltype letters not more than 1 inch and not less than 3/4 inch high. All suitable lifting and tiedown points shall be marked either "Lifting Point", "Tiedown Point", or "Lifting and Tiedown Point". Registration numbers will be furnished by the contracting officer.

3.35 Toolbox. A toolbox shall be fabricated integrally inside the cab area or a separate toolbox made of steel having a minimum nominal thickness of 0.048 inch, shall be provided. The toolbox shall have a hinged opening and a hasp accommodating a padlock shackle of 3/8 inch diameter, accessible and operable by personnel wearing gloves. The toolbox shall be bolted or welded to the crane in a protected position, shall have provisions for drainage of liquids and shall accommodate any maintenance tools furnished. The toolbox shall be treated and painted in accordance with 3.33.

3.36 Decontamination bracket assembly. A decontamination bracket assembly in accordance with MS53052 shall be furnished. The location of the bracket shall be such that it is accessible to personnel and provides clearance to remove and emplace a 5-gallon container conforming to MIL-C-1283.

3.37 Workmanship.

3.37.1 Castings and forgings. All parts, components and assemblies of the crane which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.37.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flamecutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.37.3 Fiberglass fabrication. Fiberglass reinforced plastic (FRP) and FRP coated plywood bodies shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform FRP thickness. All edges and openings of FRP coated plywood shall be completely sealed to prevent water intrusion between the laminates. The FRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the FRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

3.37.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the crane to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible, and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. The welding process used in fabrication of the crane shall be at the option of the contractor.

3.37.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by one of the following listed codes or the manufacturer's commercial code, for the type o£ welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS Dl.l, Welding Qualifications, Sections 5 and 6
Welding Qualifications of the ASME, Section IX
Earthmoving and Construction Equipment Welding, AWS/ANSI Dl4.3
Qualification, Section 5.

Contractors who only make horizontal welds need not qualify welders for "all position welding." The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The certification shall be made available for review by the contracting officer, or the contracting officer's representative.

3.37.4.2 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the crane, the crane shall

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conform to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.37.4.3 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the welding code recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and crane.

3.37.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.37.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492 or IFI Standard No. 114 for Break Mandrel Blind Rivets.

3.37.7 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and to the standards of the Materials Handling Equipment Industry or the Construction Equipment Industry.

3.38 Nuclear-biological-chemical (NBC) protection. The crane shall be operable by personnel wearing the following complete protective ensemble:

a. Chemical-biological mask, MIL-M-51282.
b. Chemical-biological hood, MIL-H-51291.
c. Chemical protective suit, MIL-S-43926.
d. Chemical protective glove set, MIL-G-43976.
e. Chemical protective footwear, MIL-F-43987

The operator shall be capable of actuating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the

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crane. The protective ensemble, as listed above, will be loaned as Government loaned equipment in order to permit performance of the tests required under test method No. 27.

3.38.1 High altitude electromagnetic pulse (EMP). Schematic and electrical diagrams detailed to the component level (i.e., resistors, capacitors, filters, transistors, SCR's) with value of all components, of the crane shall be furnished by the contractor to the Government to assist the Government in performing the EMP evaluation. A crane shall be furnished to the Government and shall be evaluated by the Government to determine necessary protection devices for protection against EMP.

3.39 Vehicle weight classification sign kit. The contractor shall apply the Government furnished vehicle weight classification numbers to each vehicle utilizing a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle. Instructions for changing the number based on the vehicle configuration shall be provided in the operator's manual.

3.40 Government-furnished equipment (GFE). The following items shall be furnished by the Government (see 6.6) for installation on the crane:

a. Vehicle weight classification sign kit, NSN 9905-00-565-6267, for each crane.
b. Case, NSN 7520-00-559-9618 (the contractor shall furnish fasteners and install the case on the crane).
c. Electrical warning decal set, NSN 7690-00-267-5778.

3.40.1 Government-loaned equipment. When contractor testing is required (see 6.2) the following items shall be loaned by the Government (see 6.6.1):

a. NBC protective ensemble for test method No. 27:

(1) Chemical-biological mask, NSN 4240-00-926-4200.
(2) Chemical-biological hood, NSN 4240-00-999-0420.
(3) Chemical protective Suit, NSN 8415-00-407-1062.
(4) Chemical protective glove set, NSN 8415-01-033-3519.
(5) Chemical protective footwear, NSN 8430-01-021-5978.

b. ISO containers.

(1) Sixteen 20-foot containers.
(2) Two 40-foot containers.

c. Spreader bars.

(1) One 20-foot spreader, NSN 3990-01-128-0089.
(2) One 40-foot spreader, NSN 3990-01-128-0091.

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d. STE/ICE for test in 4.6.2.35.1
e. DCA tester for use in 4.6.2. 35.1
f. Cold-wet gloves.

(1) Shell, NSN 8415-00-269-5700.
(2) Liner, NSN 8415-00-682-6575.

3.41 Diagnostic connector assembly (DCA). The crane shall incorporate an easily accessible DCA far interface with the Government's simplified test equipment/internal combustion Engine (STE/ICE) in accordance with STE/ICE design Guide for Vehicle Diagnostic Connector Assemblies Report No. CR-82-588-003 and MIL-T-62314, appendix B. The diagnostic connector shall be in accordance with TACOM drawing 12258941. As a minimum, the DCA shall be capable of monitoring the functions defined in table II of Report No. CR-82-588-003 and alternator/generator output voltage, alternator/generator field voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average, starter circuit resistance, battery internal resistance charge, fuel supply pressure and fuel filter restriction (Delta P). When specified (see 6.2), additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be as specified in the contract. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE vehicle test cards in the format identified in TACOM drawing 12258955 addressing both DCA and TK measurements and shall incorporate the test cards into the crane technical manuals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (preproduction) inspection (see 4.3).

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b. First article (initial production) inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 First article preproduction inspection. Prior to examination and test of the crane specified in 4.3.1.1 and 4.3.2, the following shall be performed:

a. Service the crane with oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the teats will be conducted.
b. Conduct the run-in as prescribed (4.6.2.31).
c. Set the hydraulic systems relief pressure as specified by the contractor and in 3.12 and record. Any change made to this setting shall constitute failure of all tests.
d. Furnish a schedule of maintenance to be followed during all testing of the cranes. A list of tools required to perform this maintenance shall also be provided.

4.3.1 First article (preproduction) examination.

4.3.1.1 Pretest examination. Prior to testing, all preproduction cranes shall be examined for the defects marked "X" in column 1 of table II (see 4.6.1). Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test disassembly and examination. Upon successful completion of all tests specified in 4.3.2, the crane that underwent road testing (see 4.6.2.29.3) shall be subjected to the tests marked "X" in column 3 of table III (see 4.6.2). Presence of one or more defects shall be cause for rejection.

4.3.2 First article preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, one first article crane shall be subjected to the tests marked "X" in column 1 of table III except for the functional evaluation test, load placement test and the road test. Two other first article cranes shall be subjected to the load placement and functional evaluation tests (see 4.6.2.29). A fourth first article crane shall be subjected to the road and load placement tests (see 4.6.2.29). Tests may be performed in anyorder except the DCA interface test (4.6.2.35.1) shall be conducted as the first test. Except as otherwise specified herein, test conditions shall apply throughout testing. Failure of any test except for EMP testing as specified in 4.6.2.34.1 shall be cause for rejection. Failure of a first article crane to meet any requirement specified herein shall be cause for rejection.

4.4 First article (initial production) inspection. When specified (see 3.2), four or more initial production cranes will be selected at random by the Government from the cranes being produced by production tooling and will be examined for the defects marked "X" in column 2 of table II. The cranes shall be subjected to the tests marked "X" in column 2 of table III following the same test procedure described in 4.3.2 to determine conformance to the requirements of this

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specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production crane shall not exclude the remaining cranes from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of any tests shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Examination. Each crane shall be examined for the defects marked "X" in column 3 of table II (see 4.6.1). Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crane shall be tested as specified in column 4 of table III (see 4.6.2). Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Cranes shall be examined as specified herein in accordance with table II. An (X) in the column indicates the examinations that shall be conducted.

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TABLE II. Examination schedule.

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	X	-	101. Dissimilar metals not insulated as specified.	3.3.1
X	X	-	102. Materials not as specified.	3.3
X	X	-	103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.	3.3.1.1
X	X	-	104. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the container crane.	3.3.4
X	X	X	105. Water or fluid accumulation, drain holes not provided.	3.3.1.2
X	X	-	106. Screw threads not as specified.	3.3.2
X	X	X	107. Tire material not as specified.	3.3.3
X	X	X	108. Safety provisions not as specified.	3.5
X	X	X	109. Seatbelt missing or not as specified.	3.5.1
X	X	X	110. Backup alarm missing or not as specified.	3.5.2
X	X	-	111. Human factors not as specified.	3.6
X	X	X	112. Noise hazard warning sign missing or not as specified.	3.6.1.1

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TABLE II. Examination schedule. (Continued).

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	X	-	113. Maintainability provisions not as specified.	3.7
X	X	X	114. Engine not as specified.	3.10
X	X	X	115. Starting priming system missing or not as specified.	3.10.2.1
X	X	X	116. Alternator not as specified.	3.10.2.2
X	X	X	117. Starting system not as specified.	3.10.2.3
X	X	-	118. Fuel system not as specified.	3.10.2.4
X	X	X	119. Air cleaner or restriction indicator not as specified.	3.10.2.5
X	X	-	120. Cooling system not as specified.	3.10.2.6
X	X	-	121. Fan not as specified.	3.10.2.6.1
X	X	-	122. When utilized, water pump not as specified.	3.10.2.6.2
X	X	-	123. When utilized, thermostat not as specified.	3.10.2.6.3
X	X	-	124. Radiator not as specified.	3.10.2.6.4
X	X	-	125. Gaskets, seals, and packing not as specified.	3.10.2.6.5
X	X	-	126. Engine coolant hose not as specified.	3.10.2.6.6
X	X	-	127. Governor not as specified or not set to manufacturer recommended speed.	3.10.2.7
X	X	-	128. Exhaust system not as specified.	3.10.2.8
X	X	-	129. Crankcase ventilation system not as not as specified.	3.10.2.9
X	X	-	130. Oil filters not as specified.	3.10.2.10
X	X	-	131. Tachometer drive not as specified.	3.10.2.11
X	X	X	132. Oil sampling valve or label not as specified.	3.10.2.12
X	X	-	133. Drive assembly not as specified	3.11
X	X	X	134. Transmission not as specified	3.11.1 and 3.11.1.1
X	-	-	135. Hydraulic system not as specified.	3.12
X	-	-	136. Hydraulic lines not as specified	3.12.1
X	-	-	137. pressure hose assemblies not as specified	3.12.1.1
X	-	-	138. Pressure tubing not as specified.	3.12.1.2
X	-	-	139. Pressure tube fittings not as specified.	3.12.1.3

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TABLE II. Examination schedule. (Continued).

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	-	-	140. Suction lines not as specified.	3.12.1.4
X	-	-	141. Return lines not as specified.	3.12.1.5
X	-	-	142. System filtration not as specified.	3.12.3 thru 3.12.3.2
X	-	-	143. Reservoir not as specified.	3.12.4
X	-	-	144. Relief valve not as specified.	3.12.5
X	-	-	145. Hydraulic control valve not as specified.	3.12.6
X	-	-	146. Cylinders not as specified.	3.12.7
X	X	X	147. External leakage exceeds limits specified.	3.12.9
X	-	-	148. Boom assembly not as specified.	3.13.2
X	-	-	149. Wheels not as specified.	3.15.1
X	X	-	150. Tires not as specified.	3.15.2
X	X	-	151. Wheelguards not as specified.	3.16
X	-	-	152. Steering or steering control not as specified.	3.17, 3.17.1
X	X	X	153. Brake fluid not as specified.	3.18
X	-	-	154. Service breaks not as specified.	3.18.1
X	-	-	155. Parking brakes not as specified.	3.18.2
X	X	-	156. Operator access not as specified.	3.19
			157. Seat not as specified.	
X	X	-	158. Hood not as specified.	3.20
X	X	-	159. Instrument panel not as specified.	3.21
X	X	-	160. Controls or instrumentation not as specified.	3.22
X	X	X	161. Charging system monitor not as specified.	3.23
X	X	X	162. Engine oil pressure monitor not as specified.	3.23.1
X	X	X	163. Engine coolant temperature monitor not as specified.	3.23.2
X	X	X	164. Transmission temperature monitor not as specified.	3.23.3
X	X	X	165. Air pressure monitor not as specified.	3.23.5
X	X	X	166. Hourmeter not as specified.	3.23.6
X	X	X	167. Fuel monitor not as specified.	3.23.7
X	X	X	168. Engine run control not as specified.	3.23.8
				3.23.9

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TABLE II. Examination schedule. (Continued)

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	X	X	169. Light switches not as specified.	3.23.10
X	X	X	170. Brake pedals(s) not as specified.	3.23.11
X	X	X	171. Engine speed control not as specified.	3.23.12
X	X	X	172. Transmission control not as specified.	3.23.13
X	X	X	173. Hydraulic controls not as specified.	3.23.14
X	X	-	174. Starter system not as specified.	3.23.15
X	X	-	175. Electrical system not as specified.	3.24
X	X	X	176. Batteries not as specified.	3.24.1
X	X	-	177. Battery mounting not as specified.	3.24.1.1
X	X	-	178. Battery cable and terminals not as specified.	3.24.1.2
X	X	-	179. Slave receptacles missing or not as specified.	3.24.2
X	X	-	180. Circuit breakers missing or not as specified.	3.24.3
X	X	X	181. Lights missing or not as specified.	3.24.4
X	X	X	182. Blackout lighting missing or not as specified.	3.24.4.1
X	X	X	183. Taillights missing or not as specified.	3.24.4.2
X	X	X	184. Horn missing or not as specified.	3.25
X	X	X	185. Pintle hook missing or not as specified.	3.26.1
X	X	X	186. Towing lugs missing or not as specified.	3.26.2
X	X	X	187. Personnel cab not as specified.	3.27
X	X	-	188. Glazing not as specified.	3.27.1
X	X	X	189. Cab interior not as specified.	3.27.2
X	X	X	190. Windshield, washer, or window wipers not as specified.	3.27.3
X	X	X	191. Heater missing or not as specified.	3.27.4
X	X	-	192. Lubricants not as specified.	3.29
X	X	-	193. Lubrication fittings not as specified.	3.29.1
X	X	X	194. Filling, draining, or checking provisions for lubricant enclosures missing or not as specified.	3.29.2
X	X	-	195. Fungus and moisture resistance provisions not as specified.	3.30, 3.30.1
X	X	X	196. Slinging and tiedown provisions missing or not as specified.	3.31

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TABLE II. Examination schedule. (Continued)

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	X	X	197. Identification marking missing or not as specified.	3.32
X	X	X	198. Load capacity chart missing or not as specified.	3.5.3
X	X	X	199. Instruction plate missing or not as specified.	3.32.1
X	X	X	101a. Treatment and painting not as specified.	3.33
X	X	X	102a. Lateral level indicator not as specified.	3.22
X	X	X	103a. Marking not as specified.	3.34
X	X	X	104a. Tool box not as specified.	3.35
X	X	X	105a. Decontamination unit space not as specified.	3.36
X	X	X	106a. Castings and forgings not as specified.	3.37.1
X	X	X	107a. Metal fabrication not as specified.	3.37.2
X	X	X	108a. Fiberglass fabrication not as specified.	3.37.3
X	X	X	109a. Welding and welding certification not as specified.	3.37.4 through 3.37.4.3
X	X	X	110a. Bolted connections not as specified.	3.37.5
X	X	X	111a. Riveted connections not as specified.	3.37.6
X	X	X	112a. Machine work not as specified.	3.37.7
X	-	X	113a. Schematics and electrical diagrams not as specified.	3.38.1
X	X	X	114a. Vehicle weight classification sign kit not as specified.	3.39
X	X	X	115a. Parts or components missing or do not function.	Section 3
X	X	X	116a. Assembly incorrect or not complete.	Section 3
X	X	X	117a. DCA not as specified.	3.41
X	X	X	118a. Boom length and lift angle indicating systems not as specified.	3.13.6
X	X	X	119a. Load moment indicator missing or not as specified.	3.13.7
X	X	X	120a. Crane superstructure not as specified.	3.13
X	X	X	121a. Pedestal not as specified.	3.13.1
X	X	X	122a. Main or auxiliary hook block missing or not as specified.	3.13.3

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TABLE II. Examination schedule. (Continued)

First article preproduction	First article initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5
X	X	X	123a. Main or auxiliary winch missing or not as specified.	3.13.4
X	X	X	124a. Main or auxiliary wire rope missing or not as specified.	3.13.5
X	X	X	125a. Outriggers or outrigger pads missing or not as specified.	3.14
X	X	X	126a. Rear steer indicator missing or not as specified.	3.23.4
X	X	X	127a. Interior lighting not as specified.	3.24.4.3

4.6.2 Tests. Tests shall be conducted in accordance with table III. An "X" in the column indicates the tests that shall be conducted.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. Safety and human factors considerations (see 3.5 and 3.6) shall be evaluated throughout testing. Test methods referenced in tests specified herein are those contained in the appendix. Operation of the crane shall be with fuels, oils, lubricants, and antifreeze specified herein as applicable for the temperature conditions existing at place of test.

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TABLE III. Test schedule.

First article preproduction	First article initial production	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6	7
X	X	X	X	Test conditions.	4.6.2.1	3.5, 3.6, 3.29
X	X	-	-	Engine.	4.6.2.2	3.10
X	X	-	-	Weight, dimensions, lift height, operating radii.	4.6.2.3	3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4, and 3.9.9
X	X	X	X	Lifting speed, lowering speed, lift cylinder cavitation, main and auxiliary hook block speed.	4.6.2.4	3.9.5, 3.9.6, 3.13.8
X	X	-	-	Curb clearance circle.	4.6.2.5	3.9.8
X	X	X	X	Travel speed.	4.6.2.6	3.9.3, 3.11.1
X	X	-	-	Ground clearance, and angles of approach and departure.	4.6.2.7	3.1.2.5
X	X	X	-	Drift.	4.6.2.8	3.9.10
X	X	-	-	Longitudinal gradeability, dynamic stability, emergency and parking brake.	4.6.2.9	3.9.1.2, 3.9.2, 3.18.2, 3.18.3
X	X	-	-	ANSI stability.	4.6.2.10	3.9.1.1
X	X	-	-	Steering, obstacle course.	4.6.2.11.1	3.17
X	X	-	-	Emergency steering.	4.6.2.11.2	3.17.2
X	X	-	-	Steering wheel and parking brake force.	4.6.2.11.3	3.17.1, 3.18.2
X	X	-	-	Highway & rail transportability.	4.6.2.12	3.1.3, 3.1.3.1, 3.1.3.2
X	X	-	-	Transmission stall	4.6.2.13	3.11.1, 3.26.1, 3.11, 3.10.2.8, 3.11.1.1
X	X	-	-	Hydraulic lock-up and manual over-ride.	4.6.2.17.8	3.12, 3.14
X	X	-	X	Service brake.	4.6.2.14	3.18.1
X	X	-	-	Maintainability.	4.6.2.15	3.7
X	X	-	-	Marine transportability.	4.6.2.16	3.1.3, 3.1.3.3
X	X	X	X	Leakage.	4.6.2.18	3.12.9
X	X	-	-	Hydraulic filter and filter assembly.	4.6.2.17.2	3.12.3.1, 3.12.3.2

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TABLE III. Test schedule. (Continued)

First article preproduction	First article initial production	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6	7
X	-	-	-	Pump inlet pressure.	4.6.2.17.1	3.12.1.4
X	X	X	X	Hydraulic system roll-off cleanliness.	4.6.2.17.3	3.12.8
X	X	-	-	Hydraulic cylinders.	4.6.2.17.4	3.12.7
X	-	-	-	Pump contaminant tolerance.	4.6.2.17.5	3.12.2
X	X	-	-	Hydraulic fittings	4.6.2.17.6	3.12.1.1
X	-	-	-	Hydraulic control valves.	4.6.2.17.7	3.12.6
X	X	-	-	Electromagnetic interference.	4.6.2.19	3.28
X	X	-	-	Slinging provisions.	4.6.2.20	3.31, 3.1.3.1
X	X	-	-	Tiedown provisions.	4.6.2.21	3.31, 3.1.3.1
-	-	-	-	Noise level measurement.	4.6.2.22	
X	X	-	-	Operator's station.	4.6.22.1	3.6.1.1
X	X	-	-	Exterior noise.	4.6.2.22.2	3.6.1.2, 3.25
X	X	-	-	Rain.	4.6.2.23	3.10.2.8, 3.4.4
X	X	-	-	Wheel guards.	4.6.2.24	3.16
X	X	-	-	Low temperature (-25° F)	4.6.2.25	3.4.2, 3.4.3, 3.10, 3.10.2.1, 3.10.2.6, 3.12
X	X	-	-	Wet brakes and fording. Service brake.	4.6.2.26	3.9, 3.9.4, 3.10.2.9, 3.23, 3.18.1
X	X	-	-	High speed.	4.6.2.27	3.9.12
X	X	-	-	Controls.	4.6.2.28	3.23.12, 3.23.13
X	X	-	-	Endurance.	4.6.2.29	3.9.11, 3.10.2.6, 3.10, 3.11, 3.12, 3.11.1, 3.11.1.1
X	X	-	-	Reliability.	4.6.2.29	3.8, 3.10, 3.10.2.4, 3.10.2.6, 3.11, 3.11.1.1, 3.12
X	X	-	-	Maintainability.	4.6.2.30	3.7
X	X	-	X	Run-in.	4.6.2.31	Section 3
X	X	-	-	Rear axle oscillation.	4.6.2.32	3.11.2
X	-	X	-	Post-test.	4.6.2.33	Section 3

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TABLE III. Test schedule. (Continued)

First article preproduction	First article initial production	Post-test	Quality conformance	Test	Test Paragraph	Requirement paragraph
1	2	3	4	5	6	7
X	X	-	-	Nuclear-biological-chemical.	4.6.2.34	3.38
X	X	-	-	Electromagnetic pulse.	4.6.2.34.1	3.38.1
X	X	X	X	Diagnostic connector assembly capability.	4.6.2.35	3.41
X	X	X	X	DCA-STE/ICE interface.	4.6.2.35.1	3.41
X	X	-	-	Load moment indicator.	4.6.2.36	3.13.7
X	X	-	-	Crane structure.	4.6.2.37	3.1.4, 3.13.5, 3.13.2
X	X	X	X	Boom extension speed, retraction speed, extension cylinder cavitation, simultaneous hoist and boom extension.	4.6.2.38	3.9.7, 3.13.8
X	X	-	X	Swing speed, swing brake swing lock and anti-two blocking.	4.6.2.39	3.13, 3.13.1 or 3.13.2
X	X	-	-	Heater.	4.6.2.40	3.27.4

4.6.2.2 Engine. The diesel engine shall be tested or certified in accordance with 5.2 and 5.3 of MIL-STD-1410.

4.6.2.3 Dimensions, weight, and operating radii. Determine the crane overall height, width, and axle spacing in accordance with SAE J1234. Obtain front and rear axle weights of the complete crane fueled and serviced without load with the boom fully retracted. Determine the maximum reach, and operating radii in accordance with test method No. 1. Nonconformance to 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4, and 3.9.9 shall constitute failure of this test.

4.6.2.4 Lifting speed, towering speed, lift cylinder cavitation main and auxiliary hook block speed. Test the crane in accordance with test method No. 2. Nonconformance to 3.9.5, 3.9.6, and 3.13.8 shall constitute failure of this test.

4.6.2.5 Curb clearance circle. Determine the curb clearance circle as defined in Field Test Procedure of SAE J695 with the boom fully retracted. Nonconformance to 3.9.8 shall constitute failure of this test.

4.6.2.6 Travel speed. Test the crane in accordance with test method No. 3. Speeds not as specified in 3.9.3 shall constitute failure of

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this test. Inability to shift as specified in 3.11.1 shall constitute failure of this test

4.6.2.7 Ground clearance angles of approach and departure. Test the crane in accordance with test method No. 4.
Nonconformance to 3.1.2.5 shall constitute failure of this test.

4.6.2.8 Drift. Test the crane in accordance with test method No. 5. Nonconformance to 3.9.10 shall constitute failure of this test.

4.6.2.9 Gradeability, dynamic stability, emergency and parking brake. Test the crane in accordance with test methods No. 6 and 7. Inability of the crane to operate on slopes as specified in 3.9.1.2, 3.9.2, 3.18.2, and 3.18.3 or inability of the brakes to stop and hold the crane shall constitute failure of this test.

4.6.2.10 ANSI stability. Test the crane stability in accordance with SAE J765. Nonconformance to 3.9.1.1 shall constitute failure of this test.

4.6.2.11 Steering.

4.6.2.11.1 Obstacle course. Test the crane in accordance with test method No.8. Inability of the crane to negotiate the test course or hitting of any markers shall be considered loss of operator control as specified in 3.17 and shall constitute failure of this test.

4.6.2.11.2 Emergency steering. Test the emergency steering capability of the crane in accordance with SAE J53. Inability of the crane to negotiate the test courses shall constitute failure of this test.

4.6.2.11.3 Steering wheel and parking brake force. Test the steering wheel and handbrake lever in accordance with test method No. 10. Nonconformance to 3.17.1 or 3.18.2 shall constitute failure of this test.

4.6.2.12 Highway and rail transportability. Highway transportability including disassembly times and disassembled crane dimensions if disassembly is required for highway transport shall be tested in accordance with teat method No. 11. Rail transportability shall be tested in accordance with MIL-STD-810, method 516.3, procedure VIII - Rail Impact. Nonconformance to 3.1.3, 3.1.3.1, or 3.1.3.2 shall constitute failure of this test.

4.6.2.13 Transmission stall. Test the crane in accordance with test method No. 12. Permanent deformation or damage to the pintle hook mounting provisions or crane frame (see 3.26.1), temperatures not in accordance with 3.11, or nonconformance to 3.11.1.1 shall constitute failure of this test. Exhaust back pressure not in accordance with 3.10.2.8 shall constitute failure of this test.

4.6.2.14 Service brake. Test the crane in accordance with test method Nos. 14 and 15. Stopping distance not as specified in 3.18.1 or loss of operator control shall constitute failure of this test.

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4.6.2.15 Maintainability. Test the crane in accordance with test method No. 17. Nonconformance to 3.7, as applicable, shall constitute failure of this test.

4.6.2.16 Marine transportability. Obtain the overall crane weight and width. Nonconformance to the requirement of 3.1.3.3 shall constitute failure of this test.

4.6.2.17 Hydraulic system.

4.6.2.17.1 Pump inlet pressure. Test each pump in accordance with test method No. 18. Pressure differential between atmospheric pressure and the vacuum in the line in excess of that specified in 3.12.1.4 shall constitute failure of this test.

4.6.2.17.2 Hydraulic filter and filter assembly. Test the hydraulic filter in accordance with test method No. 19 and either test to MIL-F-52723 or provide certification with supporting test data. Nonconformance to 3.12.3.1 or 3.12.3.2, as applicable, shall constitute failure of this test.

4.6.2.17.3 Hydraulic system roll-off cleanliness. Test each crane in accordance with test method No. 20. Continue cleanup procedure until the average of three consecutive counts conform to 3.12.8.

4.6.2.17.4 Hydraulic cylinders. Test each cylinder in accordance with test method No. 21. Nonconformance to 3.12.7 shall constitute failure of this test.

4.6.2.17.5 Pump contaminant tolerance. Either test the pump(s) in accordance with the procedure set up in NFPA Recommended Standard T.3.9.18-1976 or provide certification with supporting data. Report contamination analysis data in accordance with ANSI Standard B93.30M (see 3.12.2). Nonconformance to 3.12.2 shall constitute failure of this test.

4.6.2.17.6 Hydraulic fittings. Test the crane in accordance with test method No. 29. The contractor shall furnish a complete set of hose assemblies with reusable fittings for conduct of this test. Nonconformance to 3.12.1.1 shall constitute failure of this test.

4.6.2.17.7 Hydraulic control valves. Either test the hydraulic control valves in accordance with 3.12.6 or provide certification with supporting data. Nonconformance with 3.12.6 shall constitute failure of this test.

4.6.2.17.8 Hydraulic lock-up and manual over-ride. Test the crane in accordance with test method No. 13. Nonconformance to 3.12 or 3.14 shall constitute failure of this test.

4.6.2.18 External leakage. An external leakage test shall be conducted at the conclusion of the schedule of test indicated in table III. Observe and record accumulated leakage at practical

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intervals over a twelve hour period following the test(s). Leakage greater than 3.12.9 shall be cause for rejection.

4.6.2.19 Electromagnetic interference. The crane shall be tested in accordance with MIL-STD-461. Nonconformance to 3.28 shall constitute failure of this test.

4.6.2.20 Slinging provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD-209. Nonconformance to MIL-STD-209 shall constitute failure of this test (see 3.31 and 3.1.3.1).

4.6.2.21 Tiedown provisions. Test the tiedown provisions by applying the load specified in MIL-STD-209 for the time period specified in MIL-STD-209. Nonconformance to MIL-STD-209 shall constitute failure of this test (see 3.31 and 3.1.3.1).

4.6.2.22 Noise level measurement.

4.6.2.22.1 Operator's station. The noise level of the crane (excluding horns) shall not exceed 82dB(A) at operator's station during load lifting, load lowering, and high speed operations as specified in MIL-STD-1474 when tested under the following conditions: Engine shall be operated at maximum no load governed speed. All lifting and lowering operations shall be accomplished with the crane stationary, with a fully loaded 20 foot container at a 27-foot radius st specified engine speed (maximum no load governed speed), and with the lift/lower control in its wide open position. High speed test shall be accomplished at engine speed (maximum no load ground speed) in highest gear range, with boom in retracted position, over level ground at sufficient distance to insure maximum speed under specified conditions. All measurements shall be taken within 6 inches of the operator's right ear. Measurements shall be taken with doors and windows open and again with doors and windows closed. Nonconformance to 3.6.1.1 shall constitute failure of this test.

4.6.2.22.2 Exterior noise. Measure the exterior noise level in accordance with SAE J88. Nonconformance to 3.6.1.2 shall constitute failure of this test. The horn shall be operated and tested for conformance to 3.25.

4.6.2.23 Rain. Test the crane in accordance with test method No. 22. Nonconformance to 3.4.4 or 3.10.2.8 shall constitute failure of this test.

4.6.2.24 Wheel guards. Apply a vertical downward force of 300 pounds to the geometric canter of the horizontal surface, or to the uppermost point (not including supports or braces), whichever is applicable. The force shall be applied for not less than 2 minutes. Examine the guard for permanent deformation or fractures. Flexible wheel guards may be deformed during the test loading but shall return to their original shape when the load is removed. Nonconformance to 3.16 shall constitute failure of this test.

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4.6.2.25 Low temperature -25° F. Test the crane in accordance with test method No. 23. One or more of the following shall constitute failure of this test:

a. Damage to any part of the crane due to storage at -25° F (see 3 4.3).
b. Engine does not start within 5 minutes after beginning the first cranking (see 3.4.2).
c. Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of primers or priming aids (see 3.4.2).
d. Instruments and gages do not operate.
e. Battery charging system does not show positive charge.
f. With the operator wearing arctic mittens and boots:

(1) Power train cannot be engaged to the wheels.
(2) Any clutch, brake, shaft, or cylinder cannot be operated through two complete movement cycles.

g. Operator encounters difficulty in mounting to and dismounting from the operator's position on the crane.
h. Measured temperatures of engine oil (see 3.10), of engine coolant (see 3.10.2.6), or hydraulic oil (see 3.12) not within specified limits.

4.6.2.26 Wet brakes and fording. The crane shall be tested in accordance with test method No. 16. Nonconformance to 3.9.4, 3.18.1 or any of the following shall constitute failure of this test:

a. Evidence of water damage or malfunction of the torque converter, engine oil sump, transmission, transfer case, fuel tank, hydraulic reservoir, either axle, any wheel, instruments, gages (see 3.23) or electrical drivetrain components as applicable.
b. Malfunctioning of any electrical component.
c. Malfunctioning of any brake.
d. Malfunctioning of any hydraulic component.
e. Malfunctioning of the steering system.
f. Water entering the engine (see 3.10.2.9).

4.6.2.27 High speed. Test the crane in accordance with test method No. 24. Nonconformance to 3.9.12 shall constitute failure of this test.

4.6.2.28 Controls. Test the controls in accordance with test method No. 25. Nonconformance to 3.23.12 or 3.23.13 shall constitute failure of this test.

4.6.2.29 Endurance/reliability tests. Three cranes shall be tested for a total of 1240 hours. The testing shall be conducted in accordance with the test hour distribution below. The specified cycles shall be completed in not more than the hours listed.

Test	Crane 1	Crane 2	Crane 3

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Load placement test (test method No. 26 for 760 cycles)	180	180	180
Functional evaluation test (test method No. 30 for 200 cycles)	200	200	
Road test (see 4.6.2.29.3)			300

Reliability and endurance will be evaluated using these tests. More than 14 reliability failures (see 4.6.2.29.1) shall constitute failure of this test. No endurance failures (see 4.6.2.29.2) are allowed. Note: crane 3 shall undergo the road test first and the load placement test second. In addition, failure to meet the fuel tank capacity specified in 3.10.2.4, or exceeding the temperature limits specified for the cooling system (see 3.10.2.6), engine oil (see 3.10), transmission oil (see 3.11.1.1), universal joints (3.11), or hydraulic system (see 3.12) shall constitute failure of this evaluation.

4.6.2.29.1 Reliability failure. A reliability failure is defined as any unscheduled maintenance action requiring more than 15 minutes to repair. Those malfunctions resulting from accidents, noncompliance with prescribed operational or maintenance procedures, faulty test instrumentation, fail-safe devices operating as specified, and dependent secondary failures shall be considered nonrelevent failures. Any impending malfunction detected which constitutes a safety hazard to operating personnel or would cause secondary damage either to the item under test or testing instrumentation will be corrected prior to malfunction and considered as a relevant failure. Hours specified herein shall be productive clock hours excluding fueling, servicing, and downtime.

4.6.2.29.2 Endurance failure. An endurance failure is defined as any failure which results in permanent structural deformation ofthe crane or its components, or any failure which is corrected by redesign or by replacement of a failed component with a component of different configuration in order to meet specified performance requirements, or failure resulting in replacement of the engine, transmission, transfer case, hydraulic pumps or motors, hydrostatic pumps or motors, hydraulic cylinders or the drive axle.

4.6.2.29.3 Road test. The road test shall consist of travel unladed over unimproved roads (see 6.8.5) at an average speed of 9 mph.

4.6.2.30 Maintainability. Maintenance times shall be recorded throughout load placement testing in accordance with test method No. 26. A maintenance ratio of more than 0.25 as specified in 3.7 shall constitute failure of this test.

4.6.2.31 Run-in. Each crane shall be driven through a run-in for not less than 1 hour. The run-in shall include operating in all gears, operating all electrical equipment, and performing all hydraulic functions. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any component, or evidence of leakage of air, fuel, lubricant or hydraulic oil, shall be cause for

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rejection of the crane by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

4.6.2.32. Rear axle oscillation. Measure the total rear axle oscillation in accordance with test method No. 9. Nonconformance to 3.11.2 shall constitute failure of this test.

4.6.2.33 Post-test disassembly and examination. Upon completion of all endurance testing (see 4.6.2.29) complete the post test list in accordance with column 3 of table III. Measure and record the compression in each engine cylinder and dismantle the crane that underwent endurance testing sufficiently to permit detailed disassembly and visual examination of the following components:

a. Engine cylinder heads, all valves, two pistons and piston pins taken from the two cylinders having the lowest compression.
b. Torque converter and transmission.
c. Universal joints and drive shafts.
d. Differential.
e. Front axle- bearings and seals.
f. Complete brake system components.
g. Steering system cylinders, valves, and pump.
h. Hydraulic system pump, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during and after the tests.

Additional disassembly and examination of this crane shall be made at the option of the contracting officer. The crane shall be examined in accordance with table II, column 1. Record complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Each evidence of corrosion which affects the functions of the crane or evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.6.2.34 Nuclear-biological-chemical (NBC) protection. The crane shall be tested in accordance with test method No. 27. Nonconformance to 3.38 shall constitute failure of this test.

4.6.2.34.1 Electromagnetic pulse (EMP). The Government will evaluate the schematics and electrical diagrams to determine possible problem areas that may be encountered on the end item. A preproduction crane shall be subjected to non-destructive EMP testing by the Government at a Government facility.

4.6.2.35 DCA capability. The contractor shall perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the crane to insure wires and connections are free of error. After installation on the crane, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with the DCA tester (GFE)) prior to interfacing with STE/ICE as required under 4.6.2.35.1. Nonconformance to 3.41 shall constitute failure of the DCA.

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4.6.2.35.1 DCA - STE/ICE interfaces. Upon completion of inspection and tests specified in 4.6.2.35, test the DCA for all measurements specified in 3.41 using STE/ICE, After completion of first article testing, this test shall be repeated and the results compared to the first test (see 4.3.2). Nonconformance to 3.41 or failure of any componentry or wiring in the DCA harness, which results ininability to obtain the measurements previously recorded with the STE/ICE, shall constitute failure of this test.

4.6.2.36 Load moment indicator. Test the crane in accordance with test method No. 31. Nonconformance to 3.13.7 shall constitute failure of this test.

4.6.2.37 Crane structure. The crane shall be tested in accordance with SAE J1063 or certification with supporting data shall be provided by the contractor to the Government at the time of first article test to certify conformance to SAE J1063. Nonconformance to 3.1.4 shall constitute failure of this test. Certification with supporting data shall be provided to illustrate chain/cable conformance with 3.13.2 and wire rope conformance with 3.13.5. Nonconformance to 3.13.2 or 3.13.5 shall constitute failure of this test.

4.6.2.38 Boom extension speed, retraction speed, extension cylinder cavitation and simultaneous hoist and boom extension. The crane boom shall be tested in accordance with test method No. 28. Nonconformance to 3.9.7 or 3.13.8 shall constitute failure of this test.

4.6.2.39 Swing speed, swing lock, swing brake, and anti-two blocking. Test the crane in accordance with test method No. 32. Nonconformance to 3.13, 3.13.1 or 3.13.2 shall constitute failure of this test.

4.6.2.40 Heater. The heater shall be tested in accordance with SAE J1129. Nonconformance to 3.27.4 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select cranes at any time during the contract production period and subject these cranes to the examination specified in 4.6.1 and column 1 of table II, and to the tests marked "X" in column 1 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected initial production crane. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison crane to meet any requirement specified herein during and as a result of

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the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison crane(s) and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cranes yet to be produced by the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all cranes accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. Where cranes previously accepted by the Government are determined to be deficient, the contractor will negotiate with the Government to determine the method of correction.

4.8 Inspection of packaging. The preservation, packing, and marking of the crane shall be inspected as specified in appendix B of MIL-STD-162.

5. PACKAGING

5.1 Preservation, packing and marking. Each crane shall be preserved, packed and marked in accordance with the requirements of MIL-STD-162 for type I, mobile shipment, and with the exceptions and qualifications stated herein. The degree of preservation shall be level A, level B, level C, or commercial; and the degree of packing shall be level A, level B, or commercial as specified (see 6.2). Liquid cooling systems shall be preserved by filling with equal parts of water and antifreeze conforming to MIL-A-46153 for all degrees of preservation. Commercial preservation and packing shall comply with ASTM D 3951. Commercial preservation and packing shall be used only when it is known that the crane is intended for immediate use by CONUS consignees.

6. NOTES

6.1 Intended use. The crane is intended for use as a materials handling crane with a capability of unloading and loading containers from railroad cars and semitrailers. Operations also include loading and unloading slung palletized projectiles on ammunition transporters. The mobility of the crane, without payload, is intended to cover working areas where high flotation and traction characteristics are required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When first article tests are not required (see 3.2).
d. Time frame required for submission of the first-article model and number of cranes required (see 3.2).

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e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article model examination and tests, the contracting officer should specify time frame and which examination and tests will be conducted by the Government, and which examination and tests shall be conducted by the contractor (see 3.2).
f. When additional or specific information is required on plates (see 3.32).
g. Color(s) required (see 3.33).
h. Degree of preservation and of packing required (see 5.1).
i. When additional test parameters are required for STE/ICE (see 3.41).

6.3 First article. When a first article inspection is required, the items should be an initial production and a preproduction model. The first article should consist of 3 or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4. Incident report. If contractor first article testing is required the contracting officer should require the contractor to furnish a written report within 24 hours of any incidence of equipment malfunction or failure during the conduct o£ the test. As a minimum, the report should describe components and parts affected, test and operating conditions, date of incident, hourmeter reading, how detected, description of incident, and proposed corrective action taken.

6.5 Noise. When the noise level of the crane is 85dB(A) or greater, the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS Paragraph) shall be incorporated in the manuals and should include the requirement for hearing protection, the noise level of the crane, the distance from the crane at which 85dB(A) will always be met, and what operational situations or system configuration will reduce the level to 85dB(A).

6.6 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.40.

6.6.1 Government-loaned equipment. The contracting officer should arrange to loan the equipment specified in 3.40.1.

6.7 Contaminant tolerance_profile. The contracting officer should arrange to furnish the computer program for Contaminant Tolerance Profile required in 4.6.2.17.5, when requested by the contractor.

6.8 Definitions.

6.8.1 Rated load. A rated load is a load that is not more than 85 percent of the tipping load (see 6.8.4) in the least stable direction for a given radius with the crane resting on firm, level ground, with full cooling system, and with the fuel tank not less than onehalf

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full. The structural stress levels specified by SAE J1063 shall not be exceeded.

6.8.2 Operating radius. The operating radius is the measured horizontal distance from the centerline of rotation of the revolving superstructure without load to the centerline of the crane hook block with the load suspended.

6.8.3 Tipping point. The tipping point is the condition of crane loading wherein the load moment acting to overturn the crane is equal to the maximum moment of the crane available to resist overturning.

6.8.4 Tipping load. The tipping load is the force acting to unbalance a crane at the tipping point (see 6.8.3). It includes the weight of the hook block and all items suspended from the hook block.

6.8.5 Unimproved roads. An unimproved road is defined as a primitive form of unmaintained road where the land was initially cleared and leveled by tractor or grader.

6.8.6 Hoist. Hoisting a load shall be defined as lifting the load by using the winch only, i.e., the crane boom shall remain stationary.

6.8.7 Recovered materials. The term "recovered materials" means materials which have been collected from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically authorized.

6.8.8 Hook height. Hook height is the vertical distance from the ground to the bottom of the hook.

6.8.9 Mean time between failures (MTBF). MTBF is a value, within a range of the operating characteristic (OC) curve, which, if representative of the crane, provides a high probability of meeting the requirements of the prescribed test (see 4.6.2.29).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - AT
DLA - CS

Project 3950-A257

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APPENDIX

CONTAINER, CRANE, ROUGH TERRAIN, WHEEL MOUNTED, HYDRAULIC, DIESEL POWERED

10. SCOPE

10.1 The test methods contained in this appendix are to determine whether cranes procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B93.28 - Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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APPENDIX

TEST METHOD NO. 1

MAXIMUM LIFT HEIGHT - OPERATING RADII

1. TEST COURSE:

   Level, flat surface.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Loaded 40 ft. container (total weight 67,200 lbs).
   c. Loaded 20 ft. container (total weight 44,800 lbs.).
   d. Slings and spreader bars for handling 40 and 20 ft. containers.
   e. 15,000 load with sling.

3. TEST PROCEDURE:

   a. Set the crane on its outriggers in a level position.
   b. Lift the loaded 40 ft. container at a 22 ft. operating radius to the maximum vertical height in accordance with load rating chart. Measure and record the true vertical distance from the ground to the bottom of the hook.
   c. Repeat with the loaded 20 ft. container at a 27 ft. operating radius.
   d. Repeat with the 15,000 pound load at a 35 ft. operating radius using the auxiliary hook block and hoist system.
   e. Place the load center of the loaded 20 ft container 16 feet forward from the vertical plane of the foremost part of the front of the crane. With the crane on its outriggers, verify the lift capacity of the crane by hoisting the loaded 20 ft container.

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APPENDIX

TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED - LIFT CYLINDER CAVITATION -

MAIN AND AUXILIARY HOOK BLOCK SPEEDS

1. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch.
   c. Hydraulic pressure gages capable of reading below atmospheric pressure.

2. TEST PROCEDURE:

   a. Lifting and lowering speeds.

      (1) Record time in seconds required to raise the boom utilizing only the boom lift cylinders from the horizontal position (O degrees) to a boom angle of 60 degrees with the boom fully withdrawn.
      (2) Record time to lower the boom, utilizing only the boom lift cylinders at maximum speed (full-open valve) from a boom angle of 60 degrees to 0 degrees with the boom fully withdrawn. Note whether lowering speed remains constant throughout the lowering operation.
      (3) Repeat steps (1) and (2) minimum of 3 times.
      (4) Compute lifting speed and lowering speed and use the fastest rate obtained from each as the established speeds.
      (5) Repeat steps (1) through (4) except hoist the hook block (i.e., use the hoist mechanisms to lift the hook block instead of tile lift cylinders) with the boom at 60 degrees, fully withdrawn, and the hook block travel distance from ground level to maximum height. Record the travel distance.
      (6) Repeat step (5) using the auxiliary hoist system.

   b. Lift cylinder cavitation.

      (1) Install the pressure gages to read hydraulic pressure on both ends of the boom lift cylinder.
      (2) With the engine at low idle speed, and the boom fully extended, raise the boom from horizontal to maximum boom angle. Lower the boom to 0 degrees and return again to maximum boom angle not less than 10 times. Lowering shall be accomplished with control valve open 5 times and "feathered" 5 times. Observe and record any evidence of cavitation in the lift cylinder or loss of operator control. Record pressure readings. Note: Pressure readings are not required for determining

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lift cylinder cavitation for quality conformance testing purpose.

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APPENDIX

TEST METHOD NO. 3

TRAVEL SPEED

1. TEST COURSE:

The test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. TEST APPARATUS:

a. Linear measurement equipment.
b. Stopwatch or electronic timer.

3. TEST PROCEDURE:

a. Drive the unloaded crane to attain maximum speed prior to entering the measured test course. While accelerating, perform gear or range changes under full throttle.
b. Drive through the measured distance at maximum speed, recording the length of the distance and the time requiredto traverse it. The measured distance shall be not less than 88 feet.
c. Turn the crane around and repeat the test approaching the measured distance from the opposite end. Repeat for a total of six runs, three in each direction through the measured distance.

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APPENDIX

TEST METHOD NO. 4

GROUND CLEARANCE - ANGLES OF APPROACH AND DEPARTURE

1. TEST COURSE:

   Level, flat surface.

2. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Clinometer

3. TEST PROCEDURE:

   a. Measure the vertical distance from the surface to the lowest point on the vehicle, excluding wheels ant tires. This is the ground clearance.
   b. Measure the angle of approach and departure as defined in 3.1.2.5.

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APPENDIX

TEST METHOD NO. 5

DRIFT

1. TEST COURSE:

   Level, flat surface.

2. TEST APPARATUS:

   a. Temperature thermocouple and potentiometer or equivalent temperature measuring device.
   b. Clinometer.

3. TEST PROCEDURE:

   a. Install the thermocouple in the volumetric center of the hydraulic system oil reservoir. Heat the hydraulic oil by exercising hydraulic functions until the oil temperature exceeds 120° F. Maintain oil temperature above 120° F for the duration of the test.
   b. Raise the boom to a 60 degree boom angle at full boom extension. Lift rated load to a minimum height of 2 feet. Mark reference point on the lift cylinder.
   c. Hold the load for 1O minutes and record the cylinder drift on the lift cylinder.

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APPENDIX

TEST METHOD NO. 6

LONGITUDINAL GRADEABILITY - EMERGENCY STOPPING - PARKING BRAKE

1. TEST course:

   The length of the test course shall be not less than three times the wheelbase of the crane, graded to the specified slope, within ±1 percent grade. Grade resistance shall be sufficient to prevent significant wheel slip.

2. TEST APPARATUS:

   Speed indicator.

3. TEST PROCEDURE:

   a. Drive the crane forward up the slope, entering the course at a speed of not lese than 2 mph, nor more than 4 mph. Continue through the course, maintaining a speed of not less than 2 mph. Make a total of three such passes up the slope specified for ascending operation.
   b. Use the same procedure as described in a, moving forward down the slope specified for descending operation.
   c. Drive the crane forward onto the slope specified for brake operation, then stop and hold the crane with the service brakes for not less than 1 minute, with the crane in neutral gear range. Set the parking brake, release the service brake, and hold for not less than 1 minute. Release the parking brake, and continue up the slope off the course.
   d. Use the same procedure as described in c, moving forward down the slope at not less than 2 mph. Stop the crane within a distance of 20 feet by applying the service brakes at a pressure that prevents the rear wheels from leaving the ground.
   e. Position the crane at the top of and facing down the slope specified for braking operation. Shut off the engine and allow the crane to roll down the slope in neutral gear range, until a speed of 2 mph is attained. Apply the emergency brake provisions (parking or service brake as applicable) and bring the crane to a stop. Release the brake, allow the crane to begin to roll, and apply the brake again to stop the crane, and repeat for a total of six dead engine stops.

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APPENDIX

TEST METHOD NO. 7

DYNAMIC STABILITY

1. TEST COURSE:

   The length and width of the test course shall be not less than the crane curb clearance circle diameter, graded to the specified slope, within ±1 percent grade.

2. TEST PROCEDURE:

   a. Drive the crane onto the slope and perform a complete forward circle, at maximum steer angle, while on the slope.
   b. Repeat the procedure in a, steering in the opposite direction.

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APPENDIX

TEST METHOD NO. 8

STEERING

1. TEST COURSE:

The test course shall be a dry, level surface in accordance with figure A-1.

2. TEST APPARATUS:

Speed indicator.

3. TEST PROCEDURE:

Obstacle course No. 1 in accordance with figure A-1. Drive the crane forward, without load, over the test course, at governed speed in first gear.

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APPENDIX

TEST METHOD NO. 9

REAR AXLE OSCILLATION

1. TEST APPARATUS:

Linear measurement equipment.

2. TEST PROCEDURE:

a. Raise the crane, keeping the crane horizontal, to sufficient height to allow the rear axle to oscillate from stop to stop.
b. Measure at the center of the tire tread.
c. Pivot the rear axle fully clockwise. Measure and record the vertical distance that the tire traveled from horizontal.
d. Pivot the rear axle fully counterclockwise. Measure and record the vertical distance that the tire traveled from horizontal.

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APPENDIX

TEST METHOD NO. 10

STEERING WHEEL AND PARKING BRAKE FORCE

1. TEST APPARATUS:

   a. Force measurement equipment.
   b. Stopwatch.
   c. Linear measurement equipment.

2. TEST PROCEDURE:

   a. With the engine operating at low idle, determine the time, and the tangential force at the steering wheel rim, required to turn the wheel from stop to stop in both directions. Repeat 2 more times.
   b. Repeat 2a except that engine shall operate at maximum governed speed.
   c. Count the number of revolutions required to turn the wheel from stop to stop in each direction. Repeat 2 more times.
   d. Apply a minimum 300 pound force downward perpendicular to the rim of the steering wheel approximately midway between two spokes and hold for a minimum of 10 seconds.
   e. Repeat 2d except the force shall be applied upward.
   f. Apply a minimum 150 pounds force tangentially to the rim of the steering wheel, ant hold for a minimum of 10 seconds.
   g. Repeat 2f in the opposite direction.
   h. Measure seat to steering wheel clearance and directional control to steering wheel clearance.
   i. Determine the force necessary to apply the parking brake at the center of the handgrip or at the center o£ the foot pedal whichever is applicable, tangential to the arc of its travel for pivoted types, or parallel to the path of its travel for lever types. Record the highest force observed.

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APPENDIX

TEST METHOD NO. 11

HIGHWAY TRANSPORTABILITY

1. TEST APPARATUS:

   a. M747 semi-trailer(s) and suitable towing tractor.
   b. Crane of sufficient capacity to handle disassembled sections of the crane.

2. TEST PROCEDURE:

   a. Using the procedures contained in the transportability report furnished by the contractor, disassemble the crane to highway transport configuration.
   b. Record time for disassembly, equipment, and manpower utilized, and weight of each major component disassembled.
   c. Load each transport load aboard the semi-trailer(s).
   d. Record height, width, length, and total weight of each transport load.
   e. Record total time for disassembly and loading, not including time allotted for weighing.
   f. Reassemble the crane and record total time for reassembly, equipment, and manpower utilized.

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APPENDIX

TEST METHOD NO. 12

TRANSMISSION STALL

1. TEST APPARATUS:

a. Timing device.
b. Thermocouple and potentiometer or equivalent temperature measuring device.
c. Tachometer.
d. Thermometer.
e. Pressure measurement device.

2. TEST PROCEDURE:

a. Install exhaust system pressure measurement device no more than 2 inches beyond the header outlet. Record exhaust pressure during transmission stall, procedure 2 e.
b. Install thermocouple to measure oil temperature of the transmission.*
c. Thermostat shall not be blocked open.
d. With the ambient temperature not less than 25° F, operate the crane until the transmission oil outlet temperature is a minimum of 157° F.
e. Attach the crane pintle hook assembly to a fixed object to prevent forward motion of the crane by a horizontal force on the pintle hook assembly. Shift into the highest forward gear and stall the transmission for not less than 30 seconds while the engine is operating at governed speed.
f. Record initial transmission oil temperature and maximum oil temperature after 30 seconds stall, governed speed at stall, and ambient temperature.

*NOTE: Transmission oil temperatures shall be taken at the outlet from the converter for hydrodynamic transmission.

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APPENDIX

TEST METHOD NO. 13

HYDRAULIC LOCK-UP AND MANUAL OVERRIDE

1. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch.

2. TEST PROCEDURE:

   a. Raise the boom to a 60 degree boom angle and maximum boom extension with the rated load for that radius on the hook block.
   b. Shut off engine.
   c. Attempt to do the following: Lower and retract the boom, lower the load (using the hoist system), retract the boom and outriggers, and swing the superstructure.
   d. Energize manual override to lower the boom and load, retract the boom and outriggers and swing the superstructure. Record devices used, if any, method to accomplish override, and distance and time to lower the load using the hoist system.
   e. Repeat steps a, b, c, and d except measure the distance and time required to retract the boom.
   f. Repeat steps a, b, c, and d except measure the distance and time required to lower the boom.

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APPENDIX

TEST METHOD NO. 14

SERVICE BRAKE

1. TEST COURSE:

The test course shall be a dry, level, improved surface allowing no penetration, with a grade not exceeding ±1 percent.

2. TEST APPARATUS:

a. Linear measurement equipment.
b. Detonator, brake apparatus, with powder loads, or equivalent apparatus.
c. Speed indicator.

3. TEST PROCEDURE:

a. Connect the detonator to be operated by a switch directly connected to the brake pedal so that it is actuated during the first 1/4 inch of pedal travel. The service brake shall be used for this test.
b. Drive the crane over the test course. When the crane reaches the maximum crane speed, apply the brakes. Measure the distance between the point of brake pedal application and the point of stoppage. Perform the test four times in forward (two times in each direction) on the course. Observe whether the front wheels lock or any wheel loses contact with the test course. Measure the amount of fishtail.

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APPENDIX

TEST METHOD NO. 15

BRAKE FADE

1. TEST COURSE:

The test course shall be a dry, improved surface allowing no penetration, with grades not exceeding ±1 percent.

2. TEST APPARATUS:

a. Speed indicator.
b. Decelerometer.

3. TEST PROCEDURE:

Drive the crane at 10 mph and apply the service brakes to produce a deceleration rate of 8 fps per second, ±1 fps per second, until the crane has slowed to one-half the initial speed. Immediately accelerate the crane at full throttle in high gear until the travel speed reaches the initial speed. Immediately repeat the test for a total of 35 cycles. On completion of the 35th cycle, accelerate the crane at full throttle in high gear to the initial speed and immediately conduct a service brake stopping test using procedure in test method No. 14. After the brakes have completely cooled, perform the test consisting of 35 snubs and one measured stop a second time. Make no adjustments or repairs between the two test runs.

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APPENDIX

TEST METHOD No. 16

WET BRAKE AND FORDING

1. TEST COURSE:

Body of sea water to the depth specified in 3.9.4.

2. TEST PROCEDURE:

a. Drive the crane into and out of the water a minimum of 10 times. After the 10th submersion, drive the crane from the water and immediately conduct a service brake stopping test from the maximum crane speed, using the procedure specified in 3b of test method No. 14. Service brakes shall not be applied after release in the water until the test application.
b. Drain all reservoirs that were in contact with the water and note any visual evidence of water in the liquids drained from the reservoir.

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APPENDIX

TEST METHOD NO. 17

MAINTENANCE

1. TEST FACILITY:

The test facility shall consist of a shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

a. Common handtools.
b. Special tools (if furnished by the contractor with the crane).

3. TEST PROCEDURE:

a. Examine the crane for accessibility of major assemblies.
b. Examine position of drains with respect to accessibility.
c. Using the tools, demonstrate that the major assemblies are accessible for repair and maintenance.
d. Demonstrate the accessibility of drains and path of discharge of lubricants and coolants by activating drainage controls.
e. Perform and time each operation on the maintenance operations list (see 3.7).

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APPENDIX

TEST METHOD NO. 18

PUMP INLET PRESSURE

1. TEST APPARATUS:

   a. Pressure gage (vacuum).
   b. Hydraulic fittings (as required).
   c. Thermocouple.

2. TEST PROCEDURE:

   a. Install a pressure (vacuum) gage in the pump suction line immediately adjacent to the pump and a thermocouple in the hydraulic tank.
   b. Heat the oil to specified temperature.
   c. Run the pump at the maximum speed it can encounter during crane operation and record the pump-inlet pressure, and temperature.
   d. Repeat for each pump installation on the crane.

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APPENDIX

TEST METHOD NO. 19

HYDRAULIC FILTER AND FILTER ASSEMBLIES

1. TEST APPARATUS:

a. Hydraulic test stand may be used in place of on-vehicle test.
b. Pressure gages.
c. Temperature gages.
d. Flowmeter (when test stand is used).
e. Provisions for controlling oil temperature, if necessary.

2. TEST PROCEDURE:

a. Pressurize the housing to not less than the minimum proof pressure (100 psi for return line filters or 1-1/2 times maximum system pressure for pressure filters). Hold the pressure for not less than 60 seconds and examine for leaks.
b. Install a new element in the filter housing. Establish maximum flow through the filter at an oil temperature of 150° F, ±5° F. Measure and record the pressure drop across the filter assembly.
c. Plug the filter element chamber oil exit, or otherwise obstruct all flow through the filter element, leaving the outlet port open. Establish and maintain the oil temperature at 150° F, ±5° F. Increase the upstream pressure to l0 psi more than determined in b. Observe the contamination indicator. Increase upstream pressure until the indicator enters the bypass area or until bypass flow exceeds 5 percent of the maximum system flow, whichever occurs first. Observe and record this pressure (this is the bypass opening pressure). Increase pressure until the maximum system flow is attained through the bypass valve. Observe and record this pressure.

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APPENDIX

TEST METHOD NO. 20

HYDRAULIC SYSTEM ROLL-OFF CLEANLINESS

1. TEST APPATULTUS:

   Liquid automatic particle counter.

2. TEST PROCEDURE:

   a. Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
   b. Run the engine at no-load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform an inline particle count: as described in d. below.
   c. When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in d below.
   d. The procedure for determining contamination shall be as follows:

      (1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
      (2) Use any combination of on-the-crane and off-the-crane filtration for cleanup.
      (3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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APPENDIX

TEST METHOD NO. 21

CYLINDER PROOF

1. TEST APPARATUS:

   a. Hydraulic test bench or the crane capable of developing two times system working pressure.
   b. Pressure gage.

2. TEST PROCEDURE:

   a. Double acting cylinder.

      (l) Position and mechanically hold the piston at the approximate midpoint of the cylinder.
      (2) Fill rod end of the piston with oil.
      (3) With the head end port open, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for minimum of 60 seconds. Release pressure.
      (4) Fill the head end of the piston with oil.
      (5) With the rod end port open, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for minimum of 60 seconds. Release pressure.

   b. Alternate procedure for double acting cylinder

      (1) Extend the cylinder to maximum extension by filling the head end of the cylinder with oil. Apply oil pressure equal to 1.5 times the working pressure at the head port for a minimum of 60 seconds. Release pressure.
      (2) Retract the cylinder by filling the rod end of the cylinder with oil. Apply oil pressure equal to 1.5 times the working pressure at the rod end port for a minimum of 60 seconds. Release the pressure.

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APPENDIX

TEST METHOD NO. 22

RAIN

1. TEST APPARATUS:

   a. Timing device.
   b. Measuring device (U.S. Weather Bureau type preferred).
   c. Spray fixture.

2. TEST PROCEDURE:

   a. Subject the crane to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the crane at an angle of 45 degrees, ±5 degrees, with a force equivalent to a 30-mile-per-hour-wind. The simulated rainfall shall disperse uniformly over the surfaces of the crane.
   b. Expose the top and left side of the crane to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.
   c. Repeat (b) with the top and front exposed.
   d. Repeat (b) with top and right exposed.
   e. Repeat (b) with top and rear exposed.
   f. Stop the rainfall and allow the crane to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid reservoir (except cooling system) and examine for presence of water.
   g. Check electrical components, gages, cab, and the air cleaner for water entry and damage.

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APPENDIX

TEST METHOD NO. 23

LOW-TEMPERATURE TEST (-25° F)

1. TEST APPARATUS:

   a. Cold chamber.
   b. Arctic clothing.
   c. Thermocouples with potentiometer(s) or other temperature measuring device.
   d. Battery hydrometer.
   e. Wattmeter or ammeter.
   f. Voltmeter.

2. TEST PROCEDURE:

   a. Service the crane with applicable lubricants for -25° F, operation including engine oil, transmission oil, drivetrain lubricants, hydraulic oil and antifreeze.
   b. Check the specific gravity of battery electrolyte to insure full charge.
   c. Cold soak the crane to attain a stabilized temperature no warmer than -25° F. During and after this period, examine the crane for damage. Stabilization is attained when the engine coolant, fuel, lubricating oil, and battery temperatures reach -25° F, as indicated by thermocouples located as follows:

      (1) In the two center cells of each battery between the plates and at mid-depth of the electrolyte.
      (2) In the volumetric center of the hydraulic system oil reservoir.
      (3) In the engine lubricating oil gallery and in the oil sump.
      (4) In the engine coolant in the engine-cooling-jacket discharge pipe.
      (5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the crane, not less than 3 feet from each corner of the crane at engine-crankshaft level.

   d. Maintain the -25° F, plus or -5° F, temperature throughout the entire test.
   e. One operator wearing arctic clothing, including mittens, shall perform all priming and starting operations.
   f. Substitute for supplementary batteries or other electrical cranking energy shall not be used.
   g. Glowplugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the crane and normally used in starting the engine may be used.

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APPENDIX

TEST METHOD NO. 23 (Cont'd)

h. Start the crane. Starting should be accomplished within 5 minutes of the initial attempt. Smooth engine running without unnatural or continued control manipulation, and without use of primers and priming fluids, should be obtained within 15 minutes after starting.
i. When started, run engine 15 minutes. After completion of the 15-minute operation, operate the engine as follows:

(1) 10 minutes idling.
(2) 10 minutes half throttle.
(3) 5 minutes full throttle.

Perform the 25-minute cycle three times.

j. Maintain and record temperatures prior to engine starting, immediately after engine starting, and at 10 minute intervals during warm-up and cycling, at the following (but not limited to) locations on the crane:

(l) Battery.
(2) Hydraulic tank.
(3) Engine oil.
(4) Engine coolant.
(5) Chamber air.

k. During engine cycling, and while wearing arctic mittens and boots, the operator shall perform each of the following:

(l) Engage and disengage power train to the wheels in all gears.
(2) Operate through two complete movement cycles all brakes, shafts and cylinders (as permitted by space limitations in chamber).
(3) Mount to and dismount from the operator's position on the crane.

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APPENDIX

TEST METHOD NO. 24

HIGH SPEED

1. TEST COURSE:

The test course shall be a compacted surface of sufficient size to allow continuous travel at wide open throttle in highest gear range.

2. TEST APPARATUS:

a. Speed indicator.
b. Thermocouple and potentiometer or equivalent temperature measuring device.

3. TEST PROCEDURE:

a. Block thermostat in the open position.
b. Install thermocouples in the hydraulic system reservoir at the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet (if utilized), in the torque converter outlet line adjacent to the torque converter (if utilized) and engine oil gallery.
c. Record all temperatures, including ambient before starting test.
d. Drive the unloaded crane over the test course for l hour at wide open throttle in highest forward gear range. Stop the crane and record all temperatures immediately, including ambient and universal joint temperatures. Check for universal joint temperature using temperature marker.
e. In not less than 5 minutes after stopping, repeat 3d above three times, recording all temperatures immediately upon stopping, including ambient and universal joint temperatures each hour. Total operational time not less than 4 hours.

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APPENDIX

TEST METHOD NO. 25

CONTROLS

1. TEST APPARATUS:

   a. Thermocouples and potentiometer or equivalent temperature measuring device.
   b. Force measurement equipment.
   c. Linear measurement equipment.

2. TEST PROCEDURE:

   a. Install the thermocouple at the hydraulic pump(s) inlet.
   b. With an oil temperature of 160° F, to 180° F, and the engine operating at no load governed speed, measure the force required to shift each hydraulic control valve and transmission control valve at the handgrip to each position and the distance between each control.
   c. Apply 50 pounds of force to each hydraulic control valve lever in each shifting direction and maintain the force for not less than 15 seconds.
   d. Apply 100 pounds of force to each transmission control lever in each direction and maintain the force for not less than 15 seconds.

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APPENDIX

TEST METHOD NO. 26

LOAD PLACEMENT

1. TEST COURSE:

The test course shall be a level, improved surface laid out in accordance with figure A-2.

2. TEST APPARATUS:

a. Stopwatch.
b. Two loaded 40 foot containers, one fully loaded (i.e., total weight 67,200 pounds) and the other loaded for a total weight of 40,000 pounds.
c. Two loaded 20 foot ISO containers (44,800 pounds each).
d. Sling and spreader bar for 20 foot container handling.
e. Sling and spreader bar for 40 foot container handing.
f. Thermocouple and potentiometer or equivalent temperature measuring device.
g. Two 15,000 pound loads with a 15,000 pound capacity sling.

3. TEST PROCEDURE

a. Block thermostat in the open position. Install thermocouples in the hydraulic system reservoir near the pump inlet, in the transmission sump, in the transfer case sump, in the engine coolant outlet, in the torque converter outlet adjacent to the torque converter, and in the engine oil gallery. At each 30 hour interval, measure the temperatures, including ambient, at successive hourly intervals for a period of 6 hours.
b Start with the crane set up on its outriggers in the center of the test course with the sling and 20 foot spreader bar on the hook block. The two loaded 20 foot containers shall be at positions A and B (see figure A-2). Two tagmen shall be used for engaging and disengaging the spreader bar to/from the containers.
c. Begin by engaging and lifting the container at position A. Rotate 180 degrees counterclockwise (ccw) to position C maintaining a constant operating radius throughout the rotation. Lower and disengage the container. Rotate 90 degrees clockwise (cw) to position B and engage and lift the container. Rotate 180 degrees ccw to position D maintaining a constant operating radius throughout *the* rotation. Lower and disengage the container.
d. Rotate 90 degrees cw to position C, engage and lift the container. Rotate 180 degrees cw to position A (maintaining a constant radius), lower and disengage the

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APPENDIX

TEST METHOD NO. 26 (Cont'd)

container. Rotate 90 degrees cw, engage and lift the container. Rotate 180 degrees cw to position B (maintaining a constant radius), lower and disengage the container. Rotate 90 degrees cw to position A.

e. Repeat steps c and d 319 times.
f. Steps c and d constitute a cycle. Record the time required to complete each cycle as well as a total time to complete all 320 cycles.
g. Repeat steps a through f using the two 40 foot containers and the appropriate sling and spreader bar. Note: The lighter container is at the 27 foot radius and the heavier container is at the 22 foot radius.
h. Repeat steps c and d using 15,000 pound loads (with the appropriate sling) and the auxiliary winch system. On course 2 (i.e., loads at an operating radius of 35 feet) for 120 cycles. Record time required for each cycle as well as total time to complete all 120 cycles.

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APPENDIX

TEST METHOD NO. 27

NUCLEAR-BIOLOGICAL-CHEMICAL

1. TEST COURSE:

The test course shall be a level, improved surface, laid out in accordance with course 2 of figure A-2.

2. TEST APPARATUS:

a. Two 15,000 pound loads.
b. One sling for lifting the 15,000 pound loads.
c. The complete protective ensemble

3. TEST PROCEDURE:

a. While wearing the complete protective ensemble, each and every control (i.e., levers, buttons, switches, pedals, knobs and handles) shall be actuated by the operator. Record any incident of interference caused by the ensemble that is detrimental to the operation of the crane.
b. Complete 1 cycle of the load placement test, test method No. 26, 3h, with the 15,000 pound loads on course 2.
c. Record any detrimental effects the ensemble has on the operational capabilities of the crane.

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APPENDIX

TEST METHOD NO. 28

BOOM EXTENSION AND RETRACTION SPEEDS - EXTENSION

CYLINDER CAVITATION - SIMULTANEOUS HOIST AND EXTENSION

1. TEST APPARATUS:

   a. Linear measurement equipment.
   b. Stopwatch.
   c. Hydraulic pressure gages capable of reading below atmospheric pressure.

2. TEST PROCEDURE:

   a. Boom extension and retraction speeds.

      (1) Record time to extend the boom at a 60 degree boom angle from the retracted boom position to full boom extension. Note whether speed remains constant throughout extension.
      (2) Record time to retract the boom from full boom extension at a 60 degree boom angle at maximum speed (full open valve) to within 10 feet of full boom retraction. Note whether speed remains constant throughout retraction.
      (3) Repeat steps a and b a minimum of 5 times.
      (4) Compute extension and retraction speeds and use the fastest rate obtained from each as the established speeds.

   b. Extension cylinder cavitation.

      (1) Install the pressure gages to read hydraulic pressure on both ends of the boom extension cylinder.
      (2) Raise the boon to a 60 degree boom angle with the boom fully retracted.
      (3) With the engine at low idle speed, fully extend and retract the boom maintaining a 60 degree boom angle, not less than 10 times. Boom retraction shall be accomplished with control valve wide open 5 times and "feathered" 5 times. Observe and record any evidence of cavitation in the extension cylinder or loss of operator control. Record pressure readings. Note: Pressure readings are not required for determining extension cylinder cavitation for quality conformance testing purpose.
      (4) With the engine at low idle speed, fully extend and retract the boom at minimum boom angle (maximum angle below the horizontal), not less than 10 times. Boom extension shall be accomplished with control valve wide open 5 times and feathered 5 times. Observe and record any evidence of cavitation in the extension cylinder or loss of operator control.

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APPENDIX

TEST METHOD NO. 28 (Cont'd)

c. Simultaneous hoist and extension.

(1) Attach rated load to the hook block in accordance with the load rating chart for maximum boom extension at a 60 degree boom angle.
(2) Begin at full boom retraction; the crane operator shall lower the load using the hoist control and extend the boom simultaneously.

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APPENDIX

TEST METHOD NO. 29

HYDRAULIC FITTINGS

1. TEST APPARATUS:

Common handtools.

2. TEST PROCEDURE:

a. Remove one hydraulic hose assembly that has permanent-type fittings.
b. Replace the hose assembly with an equivalent field attachable fitting only to assure interchangeability.
c. Repeat a and b above until each hose assembly with permanent-type fittings is replaced with a hose assembly with reusable fittings.
d. After completing the above test reinstall the original hose assemblies on the crane.

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APPENDIX
TEST METHOD NO. 30
FUNCTIONAL EVALUATION TEST

1. TEST COURSE:

The test course shall be a level improved surface laid out in accordance with figure A-3.

2. TEST APPARATUS:

a. Stopwatch.
b. M915 tractor and M871 semitrailer.
c. Sixteen 20 foot 180 containers of which two are fully loaded (i.e., total weight of 44,800 pounds each).
d. Sling and spreader bar for 20 foot container handling.

3. TEST PROCEDURE:

a. Start with the loaded containers at positions A and B (see figure A-3), the crane and trailer in position to load the container at A onto the trailer, and the sling and spreader bar on the hook block. Two tagmen shall be used for container engaging and disengaging.
b. Load the container at A onto the trailer and move the crane and trailer adjacent to position C. Unload the container from the trailer to C. Move the crane and trailer adjacent to B. Load the container at B onto the trailer. Move the crane and trailer adjacent to D. Unload the container from the trailer to D.
c. Move the crane and trailer adjacent to C. Load the container at C onto the trailer. Move the crane and trailer adjacent to A and unload the container from the trailer to A. Move the crane and trailer adjacent to D. Load the container at D onto the trailer. Move the crane and trailer adjacent to B and unload the container from the trailer to B.
d. Move the crane and trailer adjacent to A.
e. Repeat steps b, c, and d 199 times.
f. Steps b, c, and d constitute a cycle. Record the time required to complete each cycle as well as total time to complete all 200 cycles.

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APPENDIX

TEST METHOD No. 31

LOAD MOMENT INDICATOR

1. TEST COURSE:

The test course shall be a hard, level surface with provision for anchoring the crane to prevent tipping.

2. TEST PROCEDURE:

a. Begin with thc crane set up on its outriggers.
b. Engage and lift a 15,000 pound load. Starting at a radius well within the load rating chart for that load, slowly increase the radius until the visual warning signal (amber light) of the load moment indicator is activated. Record the radius and boom angle at this point.
c. Continue to increase the radius until the visual (red light) and audible warning single of the load moment indicator is activated. Record the radius and boom angle.
d. Continue to increase the radius until the load moment indicator automatic lockout and audible alarm have been engaged. Record the radius and boom angle at this point.
e. Repeat steps b and c with 30,000, 45,000, 60,000, and 73,000 pound actual loads (i.e., load includes the weight of slings and spreaderbars used for lifting).

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APPENDIX
TEST METHOD NO. 32

SWING SPEED - SWING LOCK - SWING BRAKE ANTI-TWO BLOCKING

1. TEST APPARATUS:

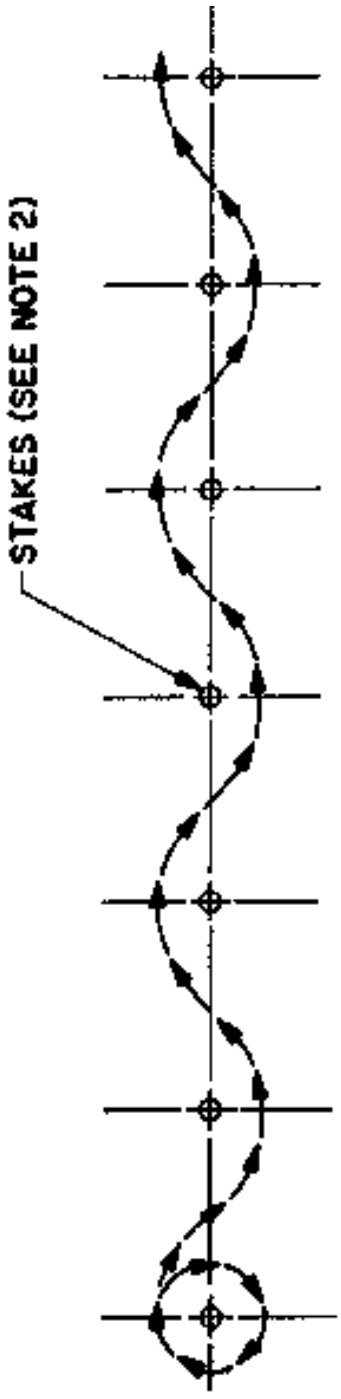
Stopwatch.

2. TEST PROCEDURE:

a. With the crane on its outriggers and level with no load, rotate the superstructure at governed engine speed with the swing control in the full open position, 360 degrees in the clockwise direction. Record the time.
b. Repeat step a 5 times.
c. Repeat steps a and b in the counter clockwise direction.
d. Engage the swing brake. With the engine at 2/3 governed speed, engage the swing control, full open, for swing in both the clockwise and counter clockwise direction. Record any rotation movement.
e. With the boom straight forward, engage the swing lock. At 2/3 governed speed, engage the swing control, full open, for swing in both the clockwise and counter clockwise direction. Record any rotation movement.
f. Set the boom angle at 60 degrees and at maximum boom length. With the hook block at ground level, hoist the hook block at governed engine speed with the hoist control at full open. Note if two-blocking occurs.

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APPENDIX



NOTES:

1. THE SPACE BETWEEN THE STAKES SHALL BE EQUAL TO TWO TIMES THE CURB CLEARANCE CIRCLE DIAMETER.
2. STAKES SHALL BE HIGH ENOUGH TO BE EASILY SEEN BY THE OPERATOR WHILE GOING THROUGH THE TEST COURSE.

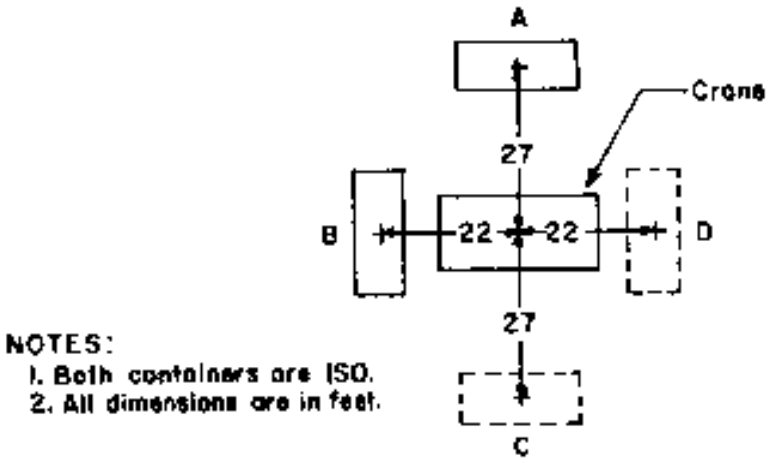
FIGURE A-1

FIGURE A-1. Obstacle course steering test.

X-2983E

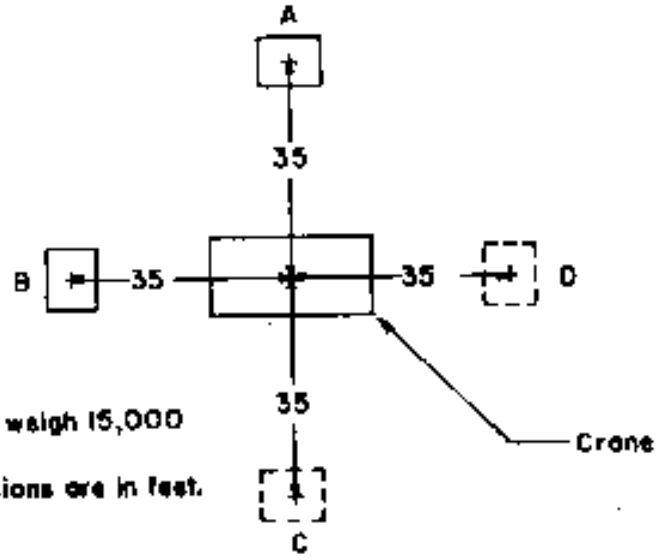
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APPENDIX



NOTES:
1. Both containers are ISO.
2. All dimensions are in feet.

Course 1



NOTES:
1. Both loads weigh 15,000 pounds.
2. All dimensions are in feet.

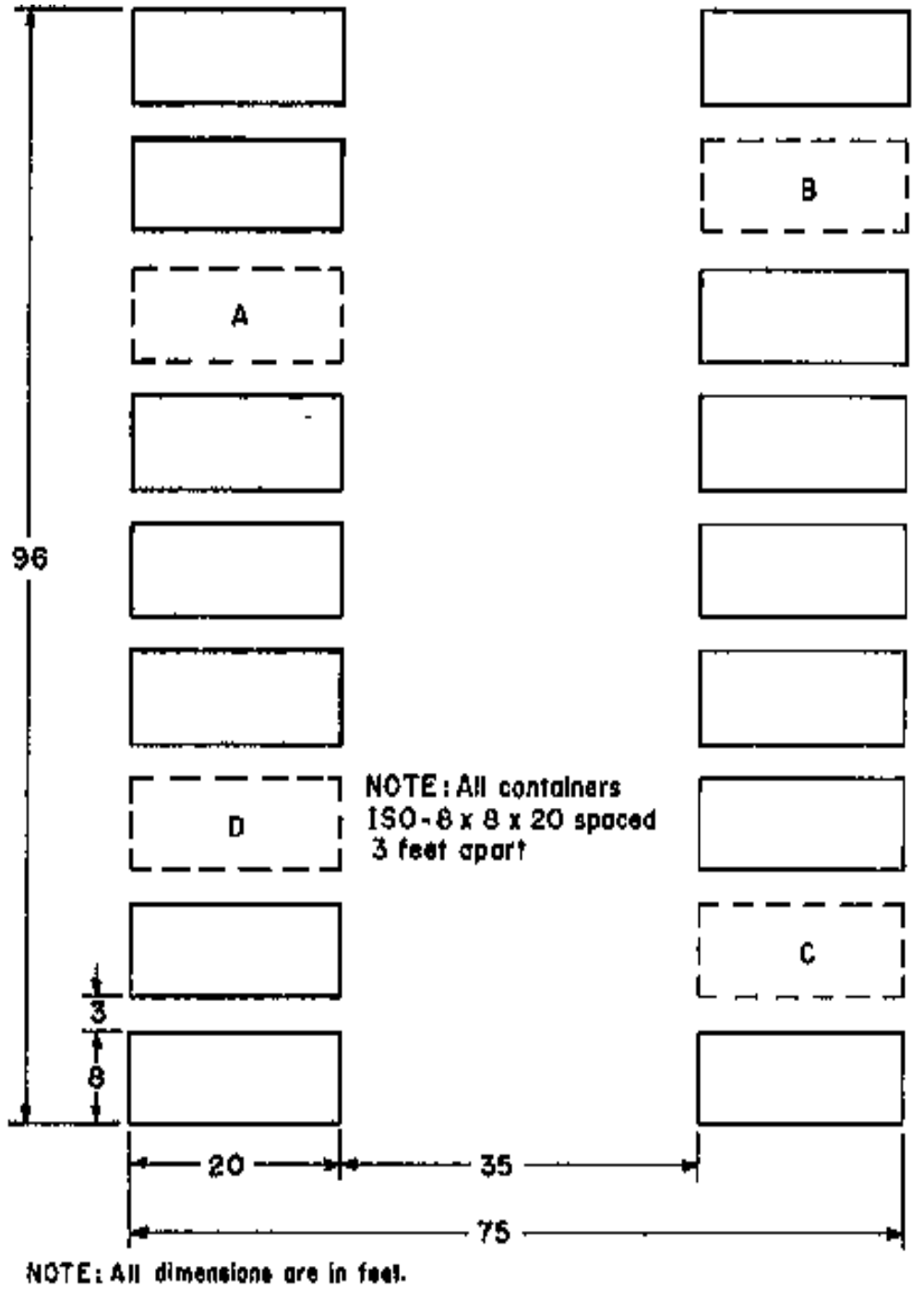
Course 2

FIGURE A-2. Load placement test.

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APPENDIX



NOTE: All dimensions are in feet.

FIGURE A-3. Functional evaluation course.

X-4318A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-53057(ME)	2. DOCUMENT DATE (YYMMDD) 850724

3. DOCUMENT TITLE
CONTAINER CRANE, ROUGH TERRAIN, WHEEL-MOUNTED, HYDRAULIC, DIESEL-POWERED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-E-ELE/21 (4EELD)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7268 (2) DSN 786-7268
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-E-ELE/21 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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